Variational AutoEncoders and Diffusion Models - Methods and Architectures

Dissertation submitted in partial fulfilment of the requirements
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Master of Technology

by

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Thesis Approval

This dissertation entitled **Variational AutoEncoders and Diffusion Models - Methods and Architectures** by **Aziz Shameem**, Roll No. 20d070020, is approved for the degree of **Master of Technology** from the Indian Institute of Technology Bombay.

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The results contained in this dissertation have not been submitted in part or in full to any other University or Institute for the award of any degree or diploma to the best of our knowledge.

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Abstract

Generative modeling has seen remarkable progress in recent years, with deep latent-variable models such as Variational Autoencoders (VAEs) and diffusion models emerging as state-of-the-art techniques for image synthesis. However, both paradigms face key challenges: VAEs often suffer from blurry outputs due to poor sample quality, while diffusion models, though capable of producing high-fidelity images, are computationally intensive, resource-hungry and relatively slow. This thesis addresses both ends of the quality–efficiency spectrum through a two-stage exploration: first by improving sample quality in VAEs using sampling-based methods, and second by redesigning diffusion architectures for enhanced computational efficiency.

In the first half of this thesis, we focus on improving image quality generated by VAEs without modifying their training process. We begin with a detailed analysis of the latent space and demonstrate that intelligent sampling—especially Langevin Monte Carlo (LMC) can substantially improve output fidelity by steering samples toward high-density regions of the learned latent distribution. We evaluate our sampling framework on the MNIST and CIFAR-10 datasets, showing significant perceptual and quantitative improvements. Additionally, we extend the method to enable conditional generation in an unsupervised VAE by using CLIP-based latent filtering. This plug-and-play approach makes post-training manipulation possible without architectural changes or retraining, adding versatility to existing models.

While these sampling strategies improve VAE outputs, they cannot bridge the fundamental expressivity gap between VAEs and modern diffusion models. Thus, the second half of this thesis shifts focus to the architecture of denoising networks within diffusion models. We study how to redesign these networks to reduce memory footprint, improve inference time, and lower parameter counts—without compromising image quality. We propose a range of architectural variants that are more resource-efficient than standard U-NET-based designs. These include: WaveUNet variants using wavelet-based downsampling and upsampling, Att2Wave which replaces expensive attention layers with lightweight WaveBlocks, InternImage-based models

leveraging deformable convolutions, and LiteVAE for latent-space diffusion. We systematically evaluate these models across CIFAR-10, MNIST, TinyImageNet, and CelebA, reporting Fréchet Inception Distance (FID) and parameter efficiency.

Experimental results highlight that models like Att2Wave and HalfWaveUNet achieve performance comparable to or better than standard U-NETs, while significantly reducing computational overhead. Particularly, Att2Wave achieves the best FID-to-parameter ratio across all datasets, demonstrating that attention mechanisms can be fully replaced by wavelet-based operations without loss in quality. The work provides concrete architectural blueprints for building fast, lightweight generative models suitable for deployment in edge devices or real-time systems.

Together, the two chapters of this thesis offer complementary solutions for resource-aware generative modeling: one through improved post-hoc sampling for VAEs, and the other through principled architectural design in diffusion models. By addressing both training-independent enhancements and core model design, this work contributes to the broader goal of making generative models more accessible, scalable, and deployment-friendly.

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Abbreviations

AE	Autoencoder
AI	Artificial Intelligence
BN	Batch Normalization
CIFAR	Canadian Institute for Advanced Research (dataset)
CLIP	Contrastive Language–Image Pretraining
CNN	Convolutional Neural Network
CVAE	Conditional Variational Autoencoder
DCN	Deformable Convolutional Network
DDPM	Denoising Diffusion Probabilistic Model
DWT	Discrete Wavelet Transform
ELBO	Evidence Lower Bound
FID	Fréchet Inception Distance
FF	Feedforward (Block)
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
IDWT	Inverse Discrete Wavelet Transform
IS	Inception Score
KL	Kullback–Leibler (Divergence)
LDM	Latent Diffusion Model
LMC	Langevin Monte Carlo
LSH	Locality Sensitive Hashing
MHA	Multi-Head Attention
ML	Machine Learning
MNIST	Modified National Institute of Standards and Technology (dataset)
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
RMSNorm	Root Mean Square Normalization

SR	Super-Resolution
t-SNE	t-distributed Stochastic Neighbor Embedding
UNET	U-shaped Convolutional Network
VAE	Variational Autoencoder
VLB	Variational Lower Bound
WaveBlock	Wavelet-Based Processing Block

Symbols

x_t	Noisy image at diffusion timestep t
x_0	Original clean image
$\hat{x}_0$	Reconstructed image from noisy input
z	Latent vector in VAE / latent-space models
$\mu(x), \Sigma(x)$	Mean and variance from encoder
ϵ_θ	Predicted noise by denoising model
ϵ	Sampled Gaussian noise
t	Diffusion timestep index
T	Total number of diffusion steps
τ	Step-size in Langevin Monte Carlo
n_0	Burn-in steps in sampling
l	Sampling interval in Markov chain
$\pi(x)$	Modified sampling distribution
μ_t, σ_t	Mean and variance in reverse process at t
$\mathcal{L}_{simple}$	Simple loss (MSE on noise)
$\mathcal{L}_{vlb}$	Variational lower bound loss
$\mathcal{X}$	Input image space
C, H, W	Channels, height, width of input image
$\mathbb{E}[\cdot]$	Expectation operator
$\mathcal{F}$	Wavelet transform operator
$\mathcal{F}^{-1}$	Inverse wavelet transform
f_f	Feedforward dimensionality (inside WaveBlock)
m	Multiplier for channel expansion (`mult`)
$D(\cdot)$	Denoising network (UNET or variant)
X_k	k^{th} sample in LMC chain
LL, LH, HL, HH	Wavelet subbands (low/high frequency splits)

Chapter 1

Understanding VAEs and Diffusion Models

1.1 Preliminaries

We start by discussing two types of generative frameworks - Variational Autoencoders (Kingma and Welling, 2022) and Denoising Diffusion Probabilistic Models (Ho et al., 2020). These two, along with Generative Adversarial Networks (Goodfellow et al., 2020) form the majority of the generative technologies currently used in production. We further highlight the relation between VAEs and DDPMs, with regards to the similarities and differences in their generation process.

1.1.1 Variational Auto Encoders (VAEs)

Autoencoders (Chen and Guo, 2023) are a type of neural network used for unsupervised learning, primarily for the purpose of dimensionality reduction or feature extraction. They consist of two main components: an encoder and a decoder. The

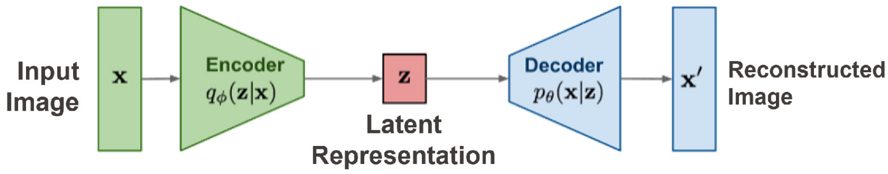

FIGURE 1.1: Autoencoders

encoder maps the input data into a compressed, lower-dimensional *latent space*, capturing its most essential features, while the decoder reconstructs the input data from this latent representation. The goal of an autoencoder is to minimize the difference between the original input and the reconstructed output (dubbed reconstruction loss), forcing the model to learn meaningful, compact representations of the data. Autoencoders are widely used in tasks like anomaly detection, image denoising, and data compression due to their ability to efficiently represent high-dimensional data in a lower-dimensional form.

Variational Autoencoders (VAEs) (Kingma and Welling, 2022) are a class of generative models that combine the power of neural networks with probabilistic inference to generate new data samples from a latent space. The key difference between VAEs and traditional autoencoders lies in how they approach the encoding process. In VAEs, the encoder network maps input data not to a single deterministic point but to a distribution (usually Gaussian) in a latent space. This is done by having the Encoder network output two quantites, mean $\mu(x)$ and variance $\Sigma(x)$, given an input point x. The latent representation is then generated by sampling a point randomly from $\mathcal{N}(\mu(x), \Sigma(x))$ (Fig 1.2). Having the decoder reconstruct the original input from a variation of points allows for generation of new, unseen data during inference time. This probabilistic framework makes VAEs particularly effective in generating smooth, continuous variations in data, which is useful in applicatfions like image

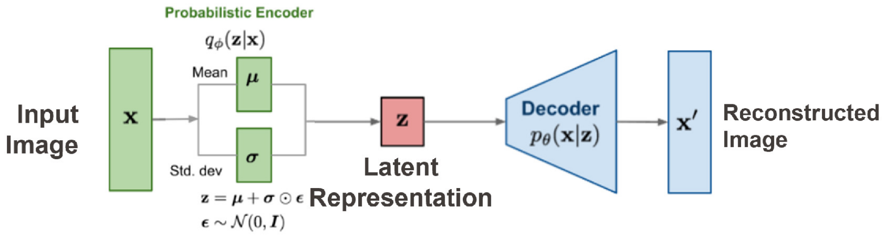

FIGURE 1.2: Variational Autoencoders

synthesis and anomaly detection.

During training, Both the models - Encoder and Decoder - and trained simultaneously using a common objective. While in the case of an Autoencoder, the objective is the simple reconstruction loss (measure of similarity between the input image and the reconstruction output by the Decoder), the objective in the case of VAEs is slightly more complex. VAEs are trained by optimizing the Evidence Lower Bound (ELBO, see appendix A), which simplifies to sum of two loss components (appendix A.1) :

- KL Loss : A loss imposed on the distibution of the latent representations. This is minimized when the distribution of the latent representations resembles the multivariate standard normal distribution.
- Reconstruction Loss : Imposed between the final reconstruction and the input datapoint, just like that in the case of an autoencoder.

During inference time in a VAE, we merely sample from the standard normal distribution $\mathcal{N}(0, I)$, and pass the samples through the decoder to obtain new generated points.

Note the the KL loss makes it so that the latent space has a known form ($\mathcal{N}(0, I)$)

that is easy to sample from. This is not the case in the traditional Autoencoders. This is what makes VAEs effective in generating new datapoints.

1.1.2 Hierarchical VAEs (HVAEs)

Hierarchical Variational Autoencoders (Kingma et al., 2016; Sønderby et al., 2016) extend the standard VAE framework by introducing multiple layers of latent variables, arranged hierarchically, to capture more complex data distributions. In a regular VAE, the model learns a single latent space that represents the data, but this can limit the model's expressiveness, especially for complex data like images or text. HVAEs address this by incorporating multiple layers of latent variables, where each layer captures different levels of abstraction in the data. The higher layers typically capture global or coarse-grained features, while the lower layers focus on fine-grained, detailed information. This hierarchical structure allows the model to better capture both the high-level structure and the intricate details of the data.

A particularly useful variant of Hierarchical VAEs is the Markovian Hierarchical VAE (MHVAEs). These introduce a specific type of dependency between the latent variables across layers. In this setup, the latent variables at each layer follow a Markov chain, where each layer's latent variable is conditioned only on the latent variable from the layer immediately above it. This Markovian assumption simplifies the model by reducing the number of dependencies, making it easier to compute and sample from. It also helps structure the information flow, with the higher layers influencing the lower layers in a sequential, controlled manner. This allows the model to gradually refine the latent representation, improving the quality of generated data and the ability to model hierarchical, structured relationships.

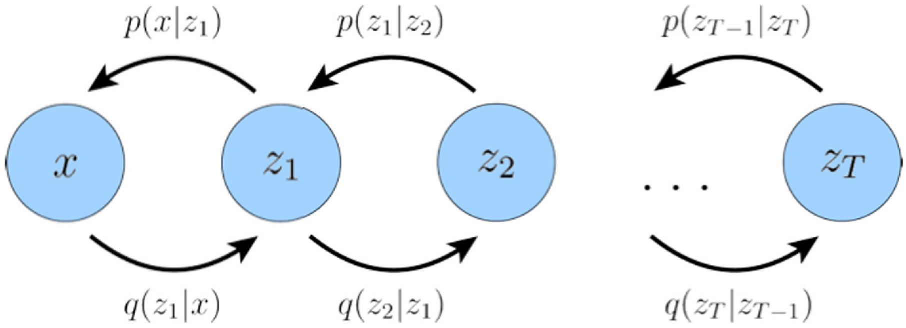

FIGURE 1.3: Markovian Hierarchical VAE

Note that both the encoding process (carried out by $q(.)$) and the decoding process (carried out by $p(.)$) are learnable, i.e. they have parameters that are learned during training. If instead we fix the encoding process so that it is devoid of any parameters, we get a more restricted mechanism of data generation. This is what we call **Diffusion Denoising Probabilistic Models**, which is the subject of the next section.

1.1.3 Denoising Diffusion Models (DDPMs)

Diffusion models (Ho et al., 2020; Nichol and Dhariwal, 2021; Bao et al., 2022; Song et al., 2021; Luo, 2022) are a class of generative models that iteratively transform a simple noise distribution into complex data structures, effectively reversing a diffusion process. The core idea behind diffusion models is to model the data generation process as a gradual refinement of random noise into meaningful samples, such as images or audio, through a sequence of denoising steps. In this process, data is progressively corrupted by adding noise over a series of time steps until it becomes indistinguishable from pure noise. The model is trained to reverse this process by learning how to denoise the noisy samples step by step, ultimately recovering the

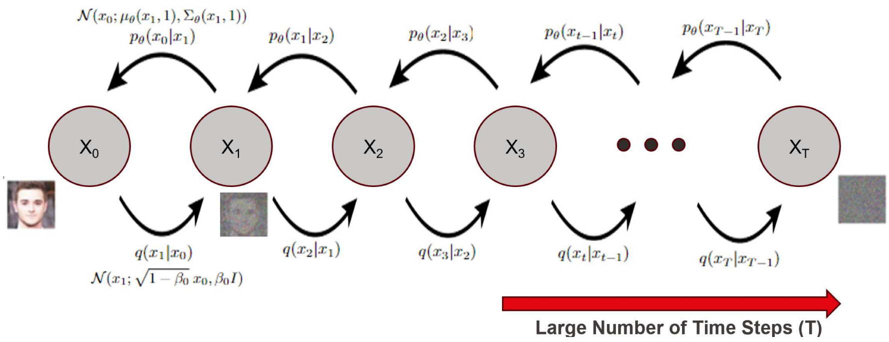

FIGURE 1.4: Diffusion Model Schematic

original data distribution.

The forward noising process is described by the equation :

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$$

Here, β_t is a sequence of hyperparameters, that is usually set to a monotonically increasing sequence in t. Thus, each step adds some amount of noise to the input. A large number of time steps, therefore, are used to simulate transformation of an input image to white gaussian noise. The reverse, learnable process of decoding is depicted by the equation

$$p_\theta(x_{t-1}|x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t); \Sigma_\theta(x_t, t))$$

Where θ are the parameters to be learned. The goal of the training is to effectively learn parameters that can perform the inverse operation to the noising process given by $q(.)$. Over the range of time steps, this can then be used to cnvert pure noise to a clean datapoint. (see fig 1.4)

Diffusion models have gained popularity due to their ability to generate high-quality data, often surpassing the results of more traditional generative models like Generative Adversarial Networks (GANs). One of the key advantages is their stability during training, as diffusion models do not suffer from the mode collapse issues that can plague GANs. Additionally, the sampling process can be controlled to trade off between quality and diversity by adjusting the number of denoising steps. This flexibility, combined with the strong theoretical foundations, makes diffusion models powerful tools for various generative tasks, including image synthesis, text-to-image generation, and audio processing.

A lot of improvements have been proposed in Diffusion Models from its inception.

As we have seen, a diffusion model is a restriction of the more general class of models, the MHVAEs, which in turn is a restriction of HVAEs. Despite this, however, diffusion models tend to outperform VAEs in the quality of image generations by a large margin, as can be seen in the latest image generation models (Rombach et al., 2022; Betker et al., 2023). This motivates the essence of this study.

The question we aim to analyze is :

Why do Diffusion Models seem to outperform Variational Autoencoders in generation tasks ?

1.2 Analysis

We begin by performing a careful analysis of the latent space of Diffusion models and VAEs, and comparing them to understand their differences.

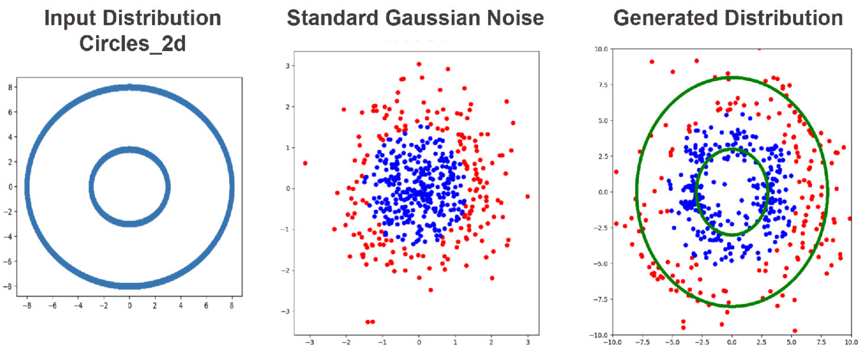

FIGURE 1.5: DDPM on Circles_2d dataset

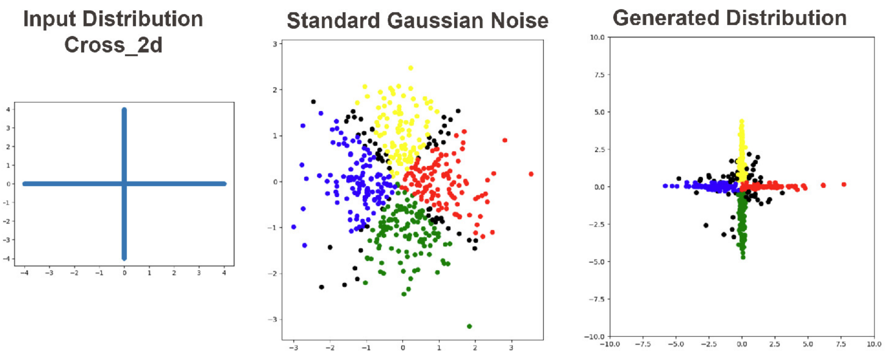

FIGURE 1.6: DDPM on Cross_2d dataset

1.2.1 Latent space in DDPMs

We perform the following experiment :

We train diffusion models on two 2-dimensional datasets (named *Circles_2d* and *Cross_2d*), shown in Figures 1.5 and 1.6. Once trained, we pass a corpus of datapoints lying in the input distributions through the trained model, and view how each point is mapped in the latent space. We color code the points to visualize the mappings, and observe a definite pattern in the results.

The observations to be noted are :

In the *Circles_2d* dataset, the points are mapped to the two circles based on their

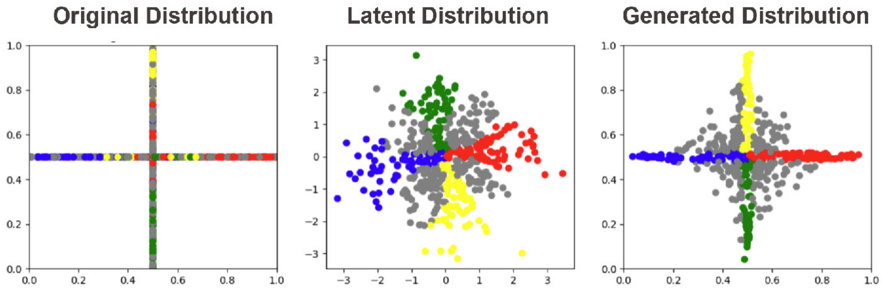

FIGURE 1.7: VAE on Cross_2d dataset

initial proximity to the circles. The points lying inside or closer to the smaller circle are mapped to it, while the ones lying outside or closer to the larger circle are mapped to the larger circle.

In the *Cross_2d* dataset as well, the points follow the proximity rule. The points are neatly mapped to the axis to which it is closest to in the input space. The ambiguous points (shown in black) are the ones that are not mapped close to any axes. These typically lie close to the lines $y = x$ and $y = -x$, which denote the areas furthest away from all axes. This agrees with the intuition that a far point that is approximately equally spaces from two modes (here, two half-axes) of the input data distribution will pose difficulties to a model that maps based on the initial proximity of the data points to the modes.

1.2.2 Latent Space in VAEs

We perform a similar experiment as in sec 1.2.1 for the case of Variational Autoencoders, In particular, we train a VAE on the *Cross_2d* dataset, and analyze the mapping produced of the input points in the latent space (Fig 1.7)

We observe a similar trend as in the case of diffusion models, with some differences :

- The datapoints are flipped and rotated : The points are flipped around the x-axis, in that those lying on the negative-y (positive-y) axis in the input space are mapped in the positive-y (negative-y) hemisphere in the latent space. Also, the rough cross-like shape in the latent space appears slightly rotated at an angle, since the topmost mode of the cross is not vertical.
  These changes can be attributed to the fact that the Encoder of the VAE is a single network (and not a series of networks that make small changes in the input), and the rotation and flipping operations can be defined by a single matrix multiplication.
- There are a lot more uncolored (grey) points in the latent space.
  The coloured areas in the latent space are much narrower in the latent space observed in the VAEs, as compared to that in Diffusion Models. These are analogous to the "Blurryness" that the generations produces by VAEs are so often attributed to.

Going one level deeper into the analysis, we conducted an experiment to study how the means and variances of the encodings of the data points were distributed. The results are shown in fig 1.8.

It can clearly be observed that the means are distributed in a perfect cross, identical to the input distribution (albeit with some scaling and rotations about the centre). However, it seems that due to the Kullback-Leibler loss in the VAE formulation, the encoder learns variances that *artificially* introduce noise in the distribution and make them spread out around the origin. However, this seems to go in a direction opposite to the desired objective (in principle, if the model is able to capture the

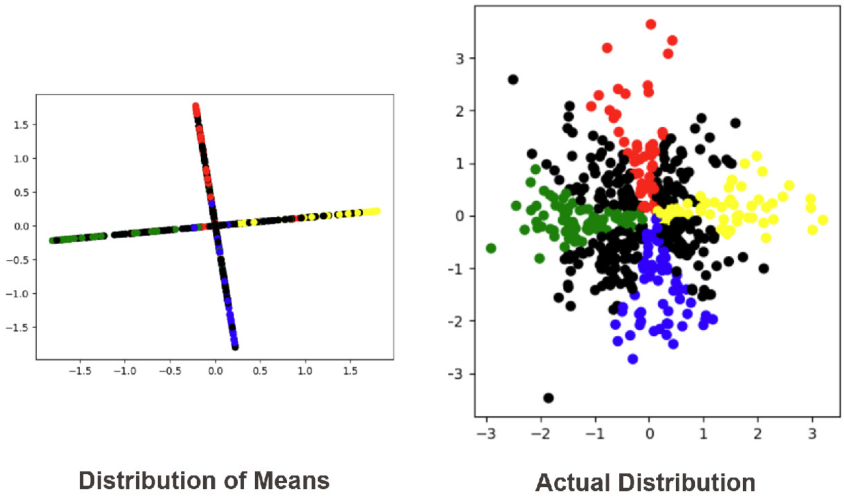

FIGURE 1.8: Analyzing the distribution of means

input distribution and transform it to another space smoothly, addition of noise to it only reduces the capacity of the decoder to faithfully reconstruct it back).

1.2.3 Custom Latent Distribution

Looking distribution of the latent representations by a VAE, we observe that the uncoloured points all seem to lie betweeen two coloured regions. Intuitively, it seems obvious that if the regions in the latent space corresponding to two disjoint modes in an input distribution overlap, the decoder will not be able to distinguish which of the modes the input point came from, given its latent representation. Noting this, it is not hard to imagine that one of the reasons for the inferior performance of the decoder in some regions (i.e. grey points) may indicate that those are the regions of overlap between the latent representation spaces of two modes (i.e. overlap between two coloured regions).

To verify the claim, we conduct the following experiment :

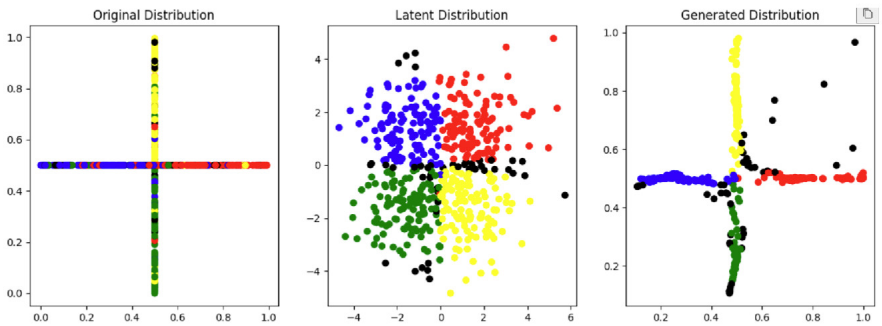

FIGURE 1.9: Custom Latent Distribution Experiment

We train a VAE with the normal objective, with the addition that the four half-axes have their latent representations mapped to the four different quadrants of the latent space, each distributed uniformly. This aims to minimize the region of overlap of the different coloured regions artificially. The results of the experiment are shown in fig 1.9. It can be seen that the distribution of the generated points is much closer to the input distribution, with the number of uncoloured (here, black) points reduced drastically. This provides some validation to the overlap hypothesis described earlier.

1.2.4 Role of the KL Loss

As another ablation on the generations produced by a VAE, we conduct an experiment where we remove the KL loss from the objective function, which now contains only the reconstruction loss. This setup resembles that of a traditional autoencder, except that the encoding process is still probabilistic in nature (the encoder learns means and variances of a normal distribution, from which it samples to obtain the latent representations). The idea behind the change is to study the role of KL Loss.

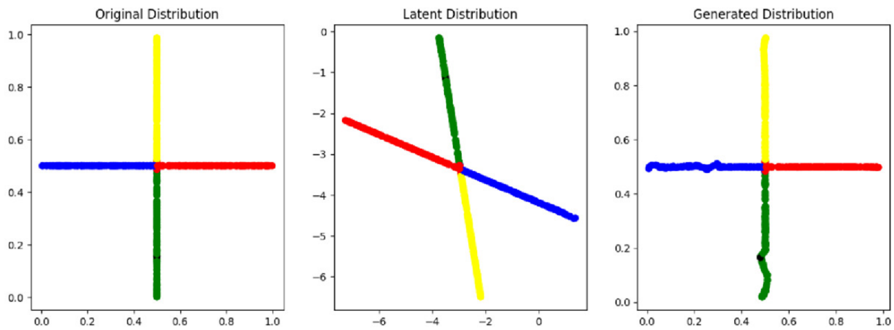

FIGURE 1.10: VAE trained without KL loss

The results of the experiment are depicted in fig 1.10. We see that the model achieves a near perfect reconstruction of the input data points (elucidated by a near absence of uncoloured points in the Generated Distribution). This also confirms that the KL loss term in the objective does not help the decoder reconstruct the representations in the latent space.

The power of the KL loss term comes up when we observe the distribution of points in the latent space. We see that this distribution is non-trivial (in this particular case, it resembles a distorted version of the distribution of the input points). However, without knowing the distribution of the latent space before hand (which is usually the case), it is not apparent how can one sample from such a distribution. The KL loss term penalizes such a situation. It pushes the latent space distribution to a standard normal distribution, which is easy to sample from.

We saw in Figure 1.8 that the two components of the VAE objective function - KL loss and reconstruction loss - seem to perform opposite functions. While the reconstruction loss aims to reduce the disparity between output of the decoder and input to the encoder (thereby enhancing the quality of generations), the KL loss only works to make latent distribution easy to sample from, and does not help the quality of generations. It follows, therefore, that if can find another way to deal with the sampling situation that does not compete with the reconstruction loss, its a win-win.

1.2.5 Latent Space Visualization Tools

Visualizing the latent space offers critical insight into the inner workings of generative models. While t-SNE and PCA are popular dimensionality reduction techniques, recent works such as UMAP (McInnes et al., 2018) have shown better cluster preservation. In our case, we use 2D scatter plots, but future analyses can benefit from tools like Plotly or TensorBoard embeddings for interactive exploration. Such visualizations help diagnose entanglement in the latent space, validate cluster purity, and aid in understanding the impact of KL regularization or sampling strategies on learned representations.

1.3 Methodology

In this section, we discuss the different methods we compare to sample from a non-standard distribution, The results obtained from the different methods are compared in section 1.4.

Please note that in the entire discussion going forward, we do not alter the training process (Kingma and Welling, 2013), nor do we change the architectures or training objectives of the Encoder or the Decoder, We merely make changes in the procedure employed during inference from a trained VAE (i.e. to obtain generations)

1.3.1 Sampling from Standard gaussian

This is the naive method that we aim to compare the others with. The points are sampled randomly from a $\mathcal{N}(0, I)$ distribution, and fed to the decoder to generate new data points.

1.3.2 Sampling using Encoding Statistics

As a first attempt, we use the statistics of the representations of the input data generated by the Encoder.

Let there be N data points in the training data, and let the latent dimension be fixed to d. The encodings of the training data can be denoted by a tensor $\alpha \in \mathbb{R}^{N\times d}$. We compute the mean $\mu(\alpha) \in$ and the covariance $\Sigma(\alpha) \in \mathbb{R}^{N\times N}$ (Juxtapose this with the mean and variance learned by the encoder of a VAE). These quantities give a description of the latent distribution. We then sample from a normal distribution with this mean and covariance, $\mathcal{N}(\mu(\alpha), \Sigma(\alpha))$.

The idea is that since the mean and covariance are taken directly from the encodings of the training data, they better describe the distribution, and thus a point sampled from such a distribution is more likely to lie in the vicinity of the training encodings. Such points will lead to better generations, since the decoder has been trained to reconstruct from similar data points.

1.3.3 Langevin Monte Carlo Sampling

Langevin Monte Carlo (Roberts and Tweedie, 1996; Dalalyan, 2014; Brosse et al., 2017; Erdogdu and Hosseinzadeh, 2020; Mousavi-Hosseini et al., 2023; Balasubramanian et al., 2022) sampling is a probabilistic technique used to draw samples from complex distributions, especially when direct sampling is challenging. It builds on the principles of both Monte Carlo methods and Langevin dynamics from physics. In this approach, a system is modeled as a particle subject to both deterministic forces (which guide it towards areas of high probability) and stochastic noise (which allows for exploration). The goal of LMC is to explore the target distribution efficiently by simulating a particle's motion under these combined forces, effectively navigating the distribution's landscape.

The core idea of LMC involves using gradient information from the log-probability distribution to guide the sampling process. Specifically, the method iteratively updates the current sample by adding a gradient ascent step, which pushes the sample towards regions of higher probability, and a small amount of Gaussian noise, which prevents the process from getting stuck in local maxima. Mathematically, this is done by updating the sample using a discretized version of the Langevin diffusion equation, ensuring that over time, the samples approximate the target distribution.

Langevin Monte Carlo is particularly valuable in high-dimensional settings where traditional Monte Carlo methods may struggle with slow convergence. By incorporating gradient information, LMC achieves faster mixing and improved sampling efficiency. It is widely used in Bayesian inference, machine learning, and variational methods where drawing samples from complex posterior distributions is necessary.

However, careful tuning of the step size and noise term is crucial to ensure that the method converges correctly and avoids biases in the sampling process.

LMC involves stepping through a markov chain, where the n^{th} state is denoted as X_n. The step equation is given by :

$$X_{k+1} = X_k + \tau \nabla \log \pi(X_k) + \sqrt{2\tau}\epsilon_k$$

There are two components of the step equation :

Gradient Ascent :

The first term in the step equation, $\tau \nabla \log \pi(X_k)$, is based on gradient ascent. Here, π is the given distribution function we intend to sample from, and this term pushes the step in the direction that causes the greatest increase in the value of the distribution function. In presence of only this term, the steps are led to a local maxima of the distribution function.

Noise Term :

The second term, $\sqrt{2\tau}\epsilon_k$, adds random noise to the step taken. here, ϵ_k is sampled from a $\mathcal{N}(0, I)$. This term causes the step to avoid getting stuck at one local maxima (where the gradient term vanishes), thereby allowing the sampling procedure to visit more than one local maxima in the course of samnpling.

Another hyperparameter apparent in the step equation if τ. This controls the magnitude of the step at each iteration. A larger value of τ cause the step size to be larger, thereby leading to greater diversity of samples obtained. However, a value too high may cause the step to lead away from the maxima of the distribution function, since the noise term will start to dominate over the step. This causes sampling from areas of low probability.

To obtain the samples, we first step through the markov chain for a pre-specified number of times, called the *Burn-in Period*, denoted here as n_0. After that, the states at an interval of l are taken as the samples.

$$\{\mathcal{S} = X_{n_0}, X_{n_0+l}, X_{n_0+2l}, \ldots\}$$

l is the interval size, another hyper parameters. A higher value of l promotes diversity in the samples.

An important part of formalizing this method to cater to the generation process of a VAE is the need to decide on a probability distribution function π. For this, we choose the following heuristic.

We want the distribution function to be such that we can reach a desired point, called *Pivot*, in one step, irrespective of the starting location. For this, we need the following to hold :

Without losing generality, we choose the *pivot* to lie at the origin (can always be done by choosing the appropriate coordinate axes). Thus, we need

$$\begin{aligned} x + \tau \nabla \log \pi &= 0 \\ \tau \nabla \log \pi &= -x \\ \log \pi &= -\frac{x^2}{2\tau} + \mathcal{C} \\ \pi &= \mathcal{A} \exp\left(\frac{-x^2}{2\tau}\right) \end{aligned}$$

A standard gaussian !

This can be generalized to the case of multiple pivots by letting the distribution function, π, be a point-wise maximum of gaussians centred around each pivot. This makes it so that only the closest pivot is influential in determining the values of the distribution function This, therefore, acts as a dynamic version of the the single pivot case for which the distribution function is derived above.

What are our pivots ? Clearly, we get the best generation when we sample an encoding of the one of the data points in the training set, since the decoder has been directly trained to reconstruct this. Thus, if we choose α to be our pivots, this ensures that the samples always lie in the vicinity of the training point encodings, and the generated points are of a higher quality.

1.3.4 Baseline

As a baseline for each of our experiment, we pass the encodings of each of the training data points, α through the decoder. Since the decoder has been directly trained on these points, the reconstructions produces by the decoder on these points is of the highest quality. This, therefore, represents the golden standard which we aim to chase in our efforts.

1.4 Results

The following section describes the experiments conducted using the above methods, and comparisons thereof.

KL loss used ?	Method	Points correctly mapped	Max distance from input distribution
No	$\mathcal{N}(\mu, \Sigma)$	480/500	0.0006
No	$\mathcal{N}(0, I)$	34/500	0.001
Yes	$\mathcal{N}(\mu, \Sigma)$	271/500	0.2717
Yes	$\mathcal{N}(0, I)$	277/500	0.3640

Table 1.1: 5d cross dataset

1.4.1 5d Cross Dataset

Firstly, we train a VAE on a 5-dimensional version of the cross dataset discussed previously. (data points lying on the half axes in a 5 dimensional space). Following this, we sample from $\mathcal{N}(0, I)$ and $\mathcal{N}(\mu, \Sigma)$ separately, and report the results in Table 1.1 The training is carried out with and without the KL loss term.

As we can see, training without the KL loss term and sampling from $\mathcal{N}(\mu, \Sigma)$ gives the best performance. Also, in both with and without KL loss, sampling from $\mathcal{N}(\mu, \Sigma)$ gives generated points that lie closer the input distribution (i.e. closer to the half axes from which the input data is taken).

1.4.2 2d Cross Dataset

We now train a VAE on the original 2-dimensional cross dataset, and sample using the three sampling methods discussed. Additionally, we report the baseline scores. See Table 1.2

Clearly, the generations become better consistently as the complexity of the sampling procesure increases. Not only is the number of correctly generated points more in langevin MC(correctness is determined by thresholding at a particular distance of the point from the closest axis), but even in the bad generations, the distance of the

Model	Method of Sampling	Number of Correct (heuristic)	Max Distance from input distribution
VAE	$\mathcal{N}(0, I)$	380/500	0.21010
VAE	$\mathcal{N}(\mu, \Sigma)$	398/500	0.14436
VAE	Langevin MC	500/500	0.04455
VAE	Input Data (baseline)	500/500	0.00800

TABLE 1.2: 2d Cross Dataset

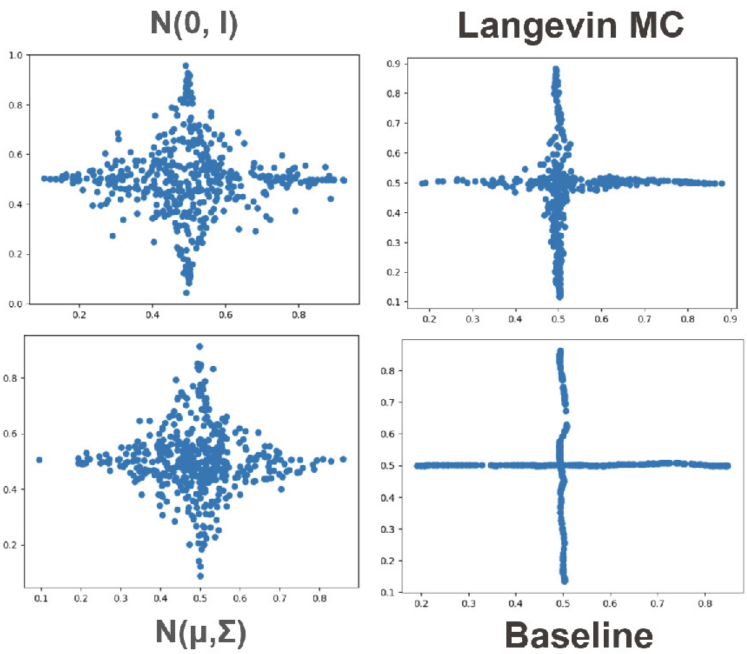

FIGURE 1.11: Results on the 2d Cross Dataset

generations to the modes of the input distribution is lesser. The qualitative results are attached in figure 1.11. The generations produces by Langevin MC sampling visually conform to the expected standards, given by the baseline.

1.4.3 MNIST Dataset

In the spirit of experimenting with increasing task complexities, we conduct the experiments on MNIST. The input datapoints are now 28x28 dimensional images. The

Model	Method of Sampling	Latent Dim	FID
VAE	$\mathcal{N}(0, I)$	2	77.8
VAE	$\mathcal{N}(\mu, \Sigma)$	2	74.5
VAE	Langevin MC	2	67.13
VAE	Input Data (baseline)	2	64.8

TABLE 1.3: Quantitative Results on the MNIST Dataset

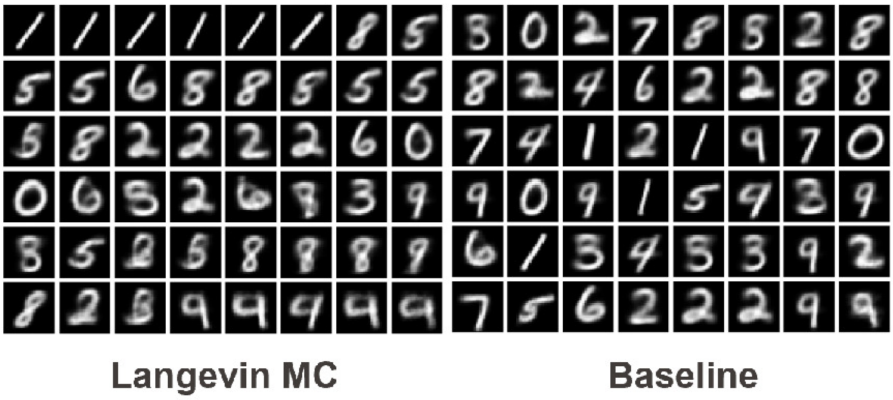

FIGURE 1.12: Qualitative Results on the MNIST Dataset

quantitative (Table 2.3) and qualitative (Figure 1.12) results are attached herewith. A similar trend as in the 2d cross dataset is observed in this as well. Qualitatively, even though the generations retain blurriness (which can be attributed to the severely constrained latent space used), the generations are very close to those in the basleine, reinforcing the benefit of altering the sampling procedure during inference.

1.4.4 Celeb-A Dataset

In the same vain, we try the methods on Celeb-A Dataset. This is a considerably complex dataset, since the inputs images have three channels (for RGB) as compared to the one channel in MNIST. As before, the qualitative (Figure 1.13) and quantitative (Table 1.4) results are shown. We observe the FID (a metric to judge quality of generations, lower is better) in this case as well decreases monotonously

Model	Method of Sampling	Latent Dim	FID
VAE	$\mathcal{N}(0, I)$	20	46.15
VAE	$\mathcal{N}(\mu, \Sigma)$	20	41.25
VAE	Langevin MC	20	40.34
VAE	Input Data (baseline)	20	39.73

TABLE 1.4: VAE Quantitative Results on the Celeb-A Dataset

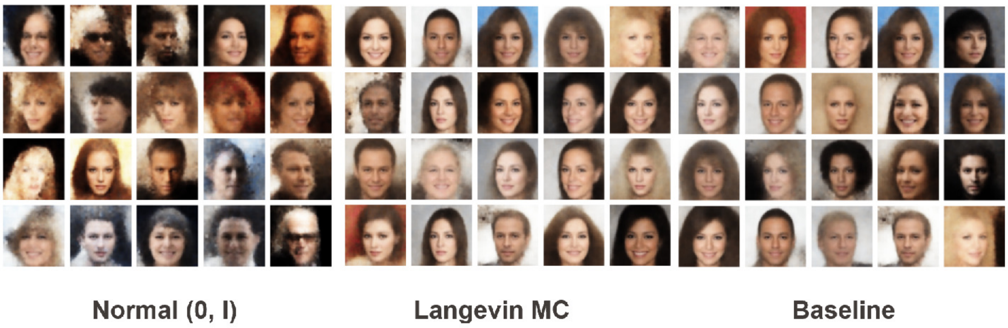

FIGURE 1.13: Qualitative Results on the Celeb-A Dataset

Model	Method of Sampling	Latent Dim	FID
AE	$\mathcal{N}(0, I)$	20	57.91
AE	$\mathcal{N}(\mu, \Sigma)$	20	51.17
AE	Langevin MC	20	50.49
AE	Input Data (baseline)	20	41.7

TABLE 1.5: AE Quantitative Results on the Celeb-A Dataset

with increase in the sophistication of the sampling process.

As an ablation, we also tried training an autoencoder (i.e. deterministic encoder-decoder) on the same dataset, the results of which are reported. In this case as well, sampling using Langevin MC sampling gives better results, although these results are not as good as the ones we obtained using VAEs.

1.5 Conditional Generation

We now extend the sampling strategy for conditional generation of images. Contrary to the traditional method of training Variational Autoencoders using labelled data, where the encoders and decoders take the condition along with the input, our method enables is to convert a VAE trained for unconditional generation to one that outputs images based on a condition, using appropriate sampling from the latent space.

Conventional Conditional VAEs (CVAEs) incorporate label information during both training and inference by conditioning the encoder and decoder on class labels. In contrast, our method retrofits conditional generation capabilities onto an already-trained unconditional VAE by modulating the sampling step. This separation is advantageous in dynamic label scenarios or multi-label settings, where explicit class-conditioning during training can be restrictive. Moreover, our method enables reusability of the trained decoder across multiple conditional setups.

1.5.1 Fixed Number of Classes

In this section, we discuss conditional generation where the classes of images are predetermined, such as MNIST (10 classes) or CIFAR (10 or 100 classes).

Firstly, we pair all the input training images with their label or set of labels. During inference, instead of considering all the encodings of training images as the pivots, we merely consider the encodings having the desired label. These pivots are then used to construct the probability distribution function π, from which points are sampled

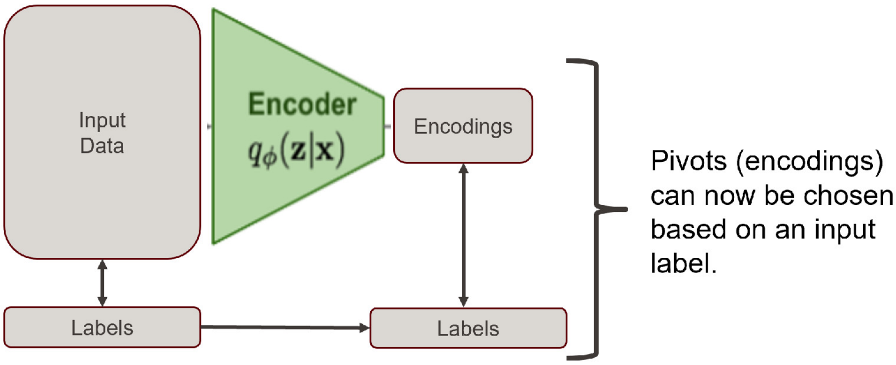

FIGURE 1.14: Conditional Generation with Fixed Number of Classes, A Schematic

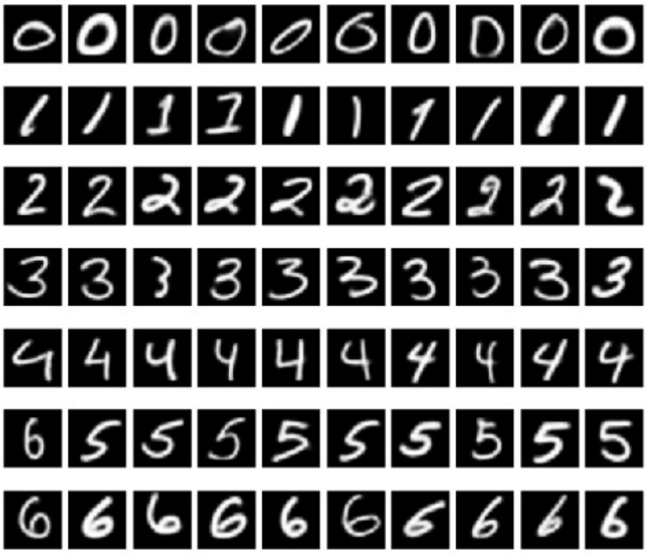

FIGURE 1.15: Example Conditional Generation with Fixed Number of Classes

(using Langevin MC sampling). This ensures that the sampled point lies close to the encodings of images belonging to the desired class, and the resulting generations, therefore, also belong to that class. The schematic of the process if shown in Fig 1.14

An important point to ensure is that the encodings of the training data should be sufficiently clustered out in order for this method to work. In high dimensional latent spaces, this is automatic (because of curse of dimensionality, all points are far apart from each other). In lower dimensional latent spaces, however, additional steps have to be taken to ensure clustering. In particular, we find that adding **contrastive loss** to the objective function functions to cluster the datapoints as desired, and leads to higher quality and correct generations by the decoder. The examples of this are attached.

Note that by nature of its formulation, this method also works in case the training images belong to more than once class. This is beneficial, since in the traditional VAEs such data points would have to be passed through the encoder multiple times corresponding to each class, potentially cause longer training times.

1.5.2 Undetermined Number of Classes

We now turn towards the case of undetermined number of classes (such as input *text* based prompts to an image generator). This is considerably more difficult than the previous section, since we need to engineer what pivots to consider before sampling and generation an output image.

We propose the following method :
In order to rank the encodings of the training images in order of their importance to the input condition, we first need to obtain a measure of their agreeability to the input prompt. This is performed using **Contrastive Language-Image Pre-Training (CLIP)**](Radford et al., 2021), which is a pretrained network consisting of an image encoder and a text encoder. The beauty of this system is that both its encoders map to the same space, and the problem of ranking the images transforms into one involving a nearest neighbour search of a query (here, the encoding of the

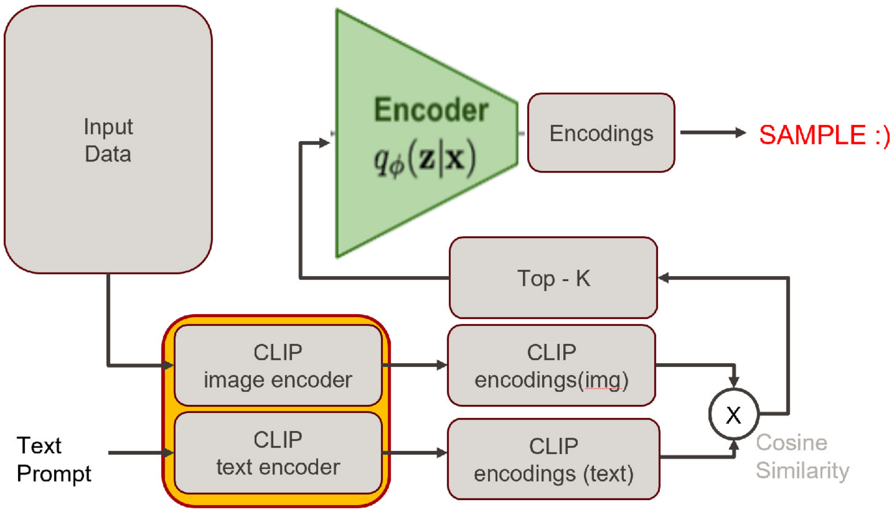

FIGURE 1.16: Conditional Generation with Undetermined Classes, A Schematic

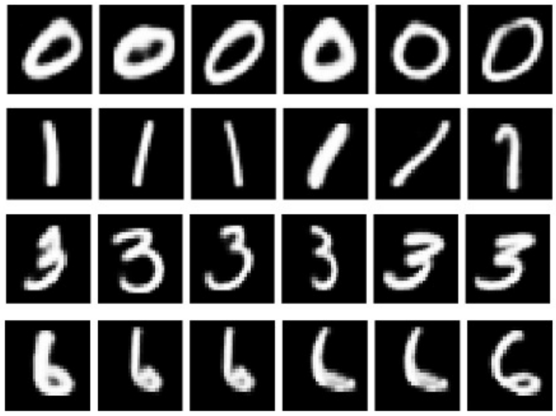

FIGURE 1.17: Example Conditional Generation with Undetermined classes

text prompt) and the corpus (here, the encodings of the input images). The complete schematic of the proposed method is shown in fig 1.16.

We first pass all the images through the image encoder (corpus), and the text prompt through the text encoder (query). Next, we compute the scores between the query and each corpus item, and store the scores. The corpus items are then sorted according to their respective scores, and the top-k (k is a hyper parameter) are taken as our *pivots* for the sampling procedure. This is then followed by generations using the trained decoder of the VAE.

1.5.3 Time Analysis and optimization

The method relies heavily on the similarity between the query and the corpus. However, computing the cosine similarity of the query with each item in the corpus is computationally very complex. The time complexity of the computation of scores is $\mathcal{O}(Nd)$, where N is the size of the corpus set and d is the latent dimension. Obtaining the top-k scores takes $\mathcal{O}(k\log N)$ time. Thus, the overall time complexity of the method becomes $\mathcal{O}(Nd)$

This is obviously too high given that we need to carry out the computation each time a generation is desired. However, what comes to the rescue is the fact that can incorporate two relaxations in the procedure :

1. We need not search for the exact top-k scores. An approximate nearest neighbour search suffices, since the representations in the latent space have the property that they are smoothly varying.

2. The procedure is very insensitive to the hyperparameter k. This means that we do not need to obtain a fixed number of pivots. Any number of pivots works, the more there are, the more diverse the generations will be, however a low number will not affect the quality or correctness of generations.

These two make it so that we can greatly reduce the time complexity of the similarity search, from $\mathcal{O}(Nd)$ to approximately $\mathcal{O}(d\,\log N)$.
We use what is known as **Locality Sensitive Hashing** for this purpose.

Locality Sensitive Hashing (LSH) (Datar et al., 2004) is a powerful technique used for approximate nearest neighbor search in high-dimensional spaces, where exact search can be computationally expensive. The core idea behind LSH is to hash input items such that similar items are more likely to be assigned to the same hash bucket, while dissimilar items are placed in different buckets. This property enables efficient similarity searches by drastically reducing the number of comparisons needed. Instead of comparing a query to every item in the dataset, LSH allows comparisons only within a small subset of items in the same or nearby hash buckets, making it ideal for large-scale data retrieval tasks like image, document, and video similarity search.

LSH relies on hash functions that are locality-sensitive, meaning they preserve the distance or similarity measure (such as cosine similarity, Euclidean distance, or Jaccard index) between data points. For instance, if two points are close to each other in the original space, they have a high probability of being mapped to the same hash value. Multiple hash functions are typically used in parallel to improve accuracy, and the results are aggregated to find the approximate nearest neighbors efficiently. This approach is particularly useful in applications where speed is critical, and exact precision can be sacrificed for a significant gain in computational efficiency, such as recommendation systems, search engines, and machine learning pipelines involving large datasets.

In our particular case, we divide the region using M planes, thereby creating 2^M partitions, called buckets. For the nearest neighbour search, we merely check which bucket the query lies in, and choose all the corpus points lying in that partition as our neighbours (or pivots). This reduces the time complexity from $\mathcal{O}(Nd)$ to

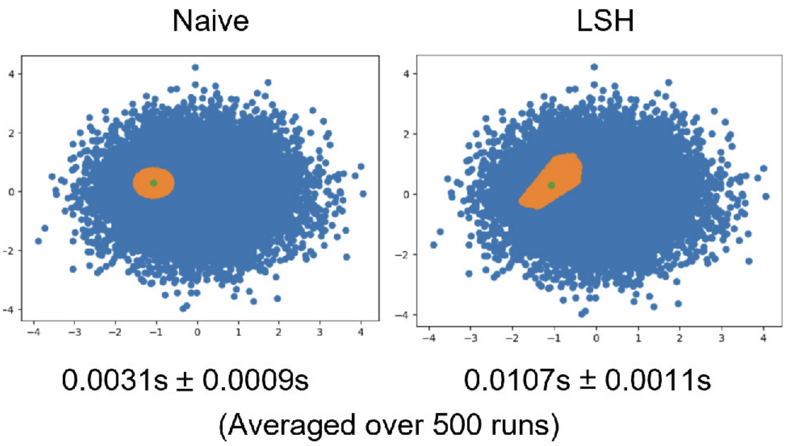

FIGURE 1.18: Time Analysis of Naive vs LSH-based Nearest Neighbour Search

The corpus points are shown in blue. The nearest neighbours identified by each method are shown in orange

$\mathcal{O}(Md)$. Since M is typically of the order of $\log N$, this causes a speedup in the computation.

In practice, the partitioning process is repeated L times, and the points lying in the correct partition in the majority of them are taken as the candidates. This causes a linear rise in time complexity, but this is still much lesser than the case where the query is compared with the entire corpus.

Figures explaining LSH, as well as generations created using this method are shown alongside.

1.6 Conclusion

In conclusion, Variational Autoencoders (VAEs) present a powerful approach to generative modeling by blending neural networks with probabilistic inference. Their structure, which involves encoding data into a latent space and subsequently reconstructing it, gives them the ability to learn meaningful, compressed representations.

Despite their strengths, VAEs often struggle with producing high-quality generations, particularly when compared to alternative generative models like diffusion models. These limitations are largely due to the interplay between the reconstruction and KL divergence losses, where the latter can introduce undesirable noise in the latent space.

Through careful experimentation, we have tried to demonstrate that by optimizing sampling strategies and modifying latent space distributions, VAEs can achieve more accurate and higher-quality data generation. Techniques such as Langevin Monte Carlo sampling provide more effective results by improving the diversity and precision of generations. Overall, the exploration of these sampling techniques suggests a promising direction for enhancing VAEs, making them more competitive for tasks that require fast and flexible data generation.

The methods we propose, if developed, boast multiple advantages of diffusion models and traditional uses of Variational Autoencoders.
Firsly, using VAEs to generate images is faster as compared to diffusion models, which typically have to iterate over a large number of time steps to generate. On the other hand, a single forward pass though the decoder of a VAE is enough for generating, and the only time spent is in the sampling process (if Langevin MC sampling is used, for example). Thus, we have greater control over the rate of generations, and this speed up increases several times when rapid and repeated generations from the same input distribution is desired.

Secondly, using the specifically created probability distribution π, we can enforce the generated images to not lie in the training data. This is a recurring problem in diffusion models, where images lying in or very close to those lying in the training data are generated. This can be consciously avoided using the sampling strategy, since we can choose each point sampled from the latent space to lie away from the latent representations of the training data.

1.7 Future Work

Future work could focus on exploring more efficient and scalable methods for both fast sampling and approximate nearest neighbor search to further enhance the performance of VAEs. One promising direction is to develop faster sampling techniques that maintain or improve the quality of generated data while significantly reducing computational costs. For example, incorporating adaptive versions of Langevin Monte Carlo or exploring Hamiltonian Monte Carlo could provide a balance between speed and precision. Additionally, optimizing the Locality Sensitive Hashing (LSH) or integrating other approximate nearest neighbor search methods, such as k-d trees or quantization-based approaches, could further accelerate the process of finding relevant points in high-dimensional latent spaces. This would be particularly useful for conditional generation tasks, where sampling from specific latent regions is required. Another avenue of research could be leveraging parallel processing or hardware acceleration (e.g., GPUs or TPUs) to speed up both sampling and nearest neighbor search, making these methods more applicable to real-world, large-scale data generation.

Chapter 2

Resource Efficient Diffusion Architectures

2.1 Introduction

Generative modeling has rapidly advanced over the past decade, with diffusion models recently establishing themselves as a state-of-the-art approach in image synthesis, outpacing traditional generative adversarial networks (GANs) in both fidelity and diversity. Originally introduced through Denoising Diffusion Probabilistic Models (DDPMs) (Ho et al., 2020), the core idea involves progressively denoising a Gaussian-corrupted input through a learned Markovian reverse process. This paradigm is theoretically grounded, sample-efficient, and produces remarkably high-quality outputs.

However, these benefits come at a steep computational cost. Unlike GANs (Goodfellow et al., 2014), which require a single forward pass, diffusion models require hundreds of denoising steps—each invoking a large neural network (typically a U-NET). As a result, they are often unsuitable for deployment in latency-sensitive or resource-constrained environments such as mobile devices, embedded hardware, or real-time generation pipelines. This presents a growing need to make diffusion models more lightweight, without compromising the high sample quality they are known for.

The architecture of the denoising network plays a pivotal role in this trade-off. Most diffusion implementations rely on U-NET backbones, often augmented with attention layers and deep residual stacks. While effective, these components contribute significantly to the overall model size, memory footprint, and inference time. The architectural design thus becomes a key bottleneck and an important axis for optimization.

This thesis investigates the question: *Can we redesign the architectural components of diffusion models to achieve comparable generative quality with significantly fewer parameters?* To this end, we explore several classes of architectures—including wavelet-based U-NET variants, attention-free models (Att2Wave), deformable convolution-based InternImage blocks, and latent-space diffusion (LiteVAE). Each design is motivated by a different hypothesis about where redundancy or inefficiency lies in the standard pipeline.

By systematically evaluating these variants on benchmark datasets such as CIFAR-10, MNIST, TinyImageNet, and CelebA, we aim to identify patterns and principles that inform future designs of efficient generative models. Our findings not only demonstrate the viability of non-attention-based and frequency-aware alternatives but also provide actionable insights into trade-offs between performance, model complexity, and hardware feasibility.

2.2 Preliminaries

2.2.1 Diffusion Model Basics

Diffusion models (Ho et al., 2020; Nichol and Dhariwal, 2021; Bao et al., 2022; Song et al., 2021; Luo, 2022) have become one of the most prominent classes of generative models in recent years. They operate on a unique principle of transforming data into noise and then learning to reverse that process. Specifically, they consist of a forward process, where Gaussian noise is gradually added to the data over a sequence of steps, and a reverse process, where a neural network is trained to progressively denoise this input. Unlike adversarial training in GANs, diffusion models rely on maximum likelihood estimation, which provides more stable training dynamics and interpretable probabilistic foundations.

At each step of the reverse process, the neural network predicts either the original image, the noise added at that step, or some intermediate representation, depending on the formulation (e.g., DDPM, DDIM (Song et al., 2022), or score-based methods (Song et al., 2020)). The denoising is performed iteratively, and despite the typically high number of steps (often 50–1000), the process can be accelerated through improvements in the underlying network or by using fewer diffusion steps. These models are particularly known for their ability to generate high-quality and diverse images, achieving state-of-the-art results in various image generation benchmarks.

However, a critical drawback of diffusion models is their computational inefficiency. Each sample requires multiple forward passes through a large neural network, which in many cases is based on the U-NET architecture. This not only results in high latency during inference but also places a substantial burden on memory and compute resources. Hence, while diffusion models outperform GANs in many qualitative and quantitative metrics, their practical adoption, especially in real-time or resource-constrained settings, is limited. Addressing these limitations is the core motivation of this work.

2.2.2 Fréchet Inception Distance (FID)

Evaluating generative models is a non-trivial task, especially because visual quality and diversity are both essential but subjective attributes. Among various evaluation metrics, the Fréchet Inception Distance (FID) proposed by Heusel et al. (2017) has emerged as the de facto standard for assessing the performance of image generation models. FID compares the statistical properties of real and generated images by passing them through a pre-trained Inception-v3 network and computing the distance between the resulting feature distributions. This is done under the assumption that these feature representations follow a multivariate Gaussian distribution.

The FID score captures both the mean and covariance of the real and generated distributions, making it sensitive to both image fidelity and diversity. Lower FID values indicate that the synthetic images are more similar to the real images in terms of high-level features. This aligns well with human visual assessment and provides a reliable quantitative metric for comparing different generative approaches. It

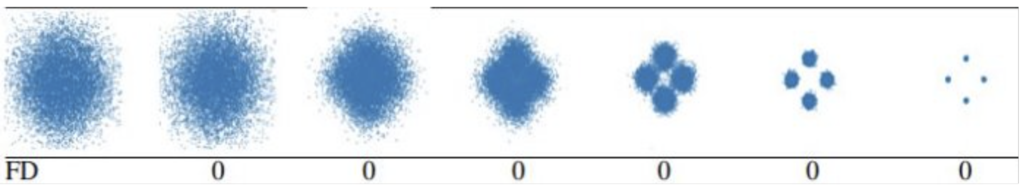

FIGURE 2.1: Different Distributions exhibiting zero Fréchet Distance(FD)
source : Pranava et. al., "Normalizing Flow Based Metric for Image Generation"

is commonly used in benchmarks such as CIFAR-10, CelebA, and ImageNet subsets.

Despite its popularity, FID is not without flaws (Stein et al., 2023; Jeevan et al., 2024a). Firstly, it relies heavily on the feature representations of an Inception-v3 model trained on ImageNet, which may not generalize well to domains like medical imaging or satellite photography. Secondly, it requires a large number of image samples—often in the tens of thousands—for a stable and accurate estimation. Lastly, FID is not sensitive to minor image distortions such as subtle noise, blur, or color shifts, and it assumes Gaussianity of feature distributions, which may not hold in practice. For these reasons, while FID remains a useful proxy, researchers often complement it with qualitative analysis and domain-specific metrics.

As an example, in Figure 2.1, different multimodal gaussian distribution sharing the same global mean and variance show zero Fréchet Distance from each other, despite clearly belonging to different distributions. Such differences in distributions, therefore, are not captured by FID when used as a metric to evaluate generative models.

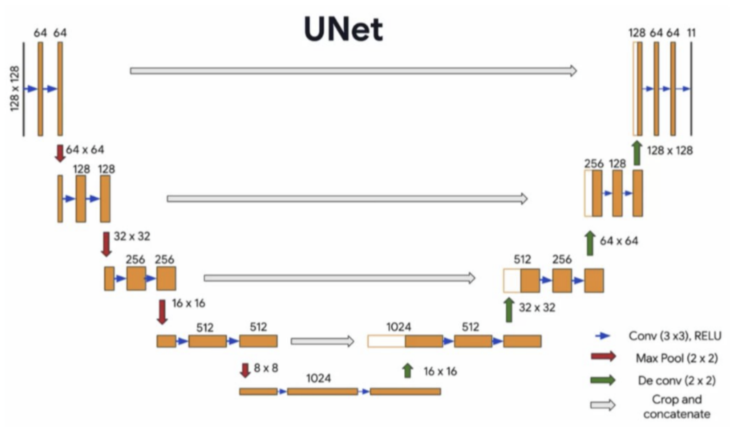

FIGURE 2.2: U-NET Architecture

2.2.3 U-NET Architecture

The U-NET architecture (Ronneberger et al., 2015) has been a foundational building block in many image processing tasks, originally designed for biomedical image segmentation. It features a symmetric encoder-decoder design, with skip connections bridging corresponding levels of the encoder and decoder. These skip connections help preserve spatial information lost during downsampling, making U-NET especially effective in tasks requiring pixel-wise reconstruction, such as segmentation and denoising. In diffusion models, U-NET is used as the denoising backbone that estimates the clean image (or the noise) at each diffusion step.

The architectural complexity of U-NET used in diffusion models (Fig. 2.3) stems from its deep residual blocks, attention layers, and repeated use of high-resolution feature maps. In its diffusion model incarnation, these blocks are parameter-intensive: attention blocks alone can have over 1.4 million parameters, while residual blocks contribute 200,000–700,000 parameters depending on their depth and width. This contributes to the overall bulkiness of the model, making it computationally expensive to train and deploy. The large memory footprint also restricts its use in

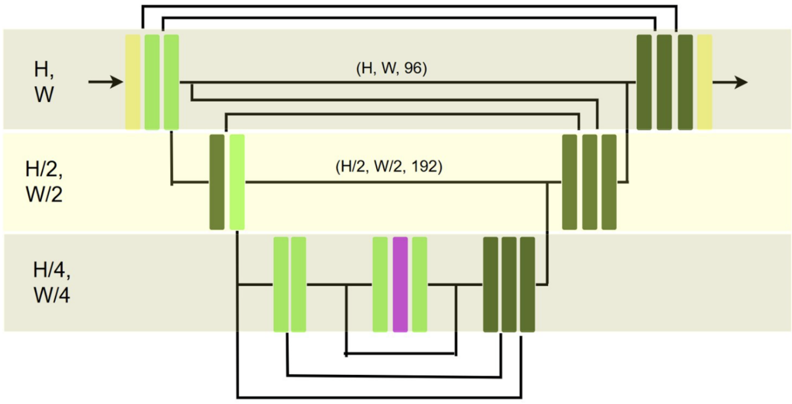

FIGURE 2.3: U-NET for Diffusion
Contains Conv(yellow), Residual(green) and Attention(magenta) blocks

lower-resource environments such as mobile or embedded devices.

Given these limitations, the focus of recent research has shifted toward modifying or replacing components of U-NET to improve efficiency. Alternatives such as lightweight convolutions, efficient sampling methods, and new spatial mixing techniques have been proposed. In this work, we explore such modifications extensively, including the use of WaveMix-based operations (Jeevan et al., 2024c, 2023, 2024b) in place of traditional convolutional or attention-based blocks. These changes aim to retain the strengths of U-NET—namely, its spatial preservation and reconstruction fidelity—while dramatically reducing its parameter count and memory requirements.

2.2.4 WaveMix

WaveMix (Jeevan et al., 2024c) is a novel neural network architecture that incorporates principles from wavelet signal processing to achieve efficient and effective feature mixing. Unlike traditional convolutional or attention-based networks, WaveMix

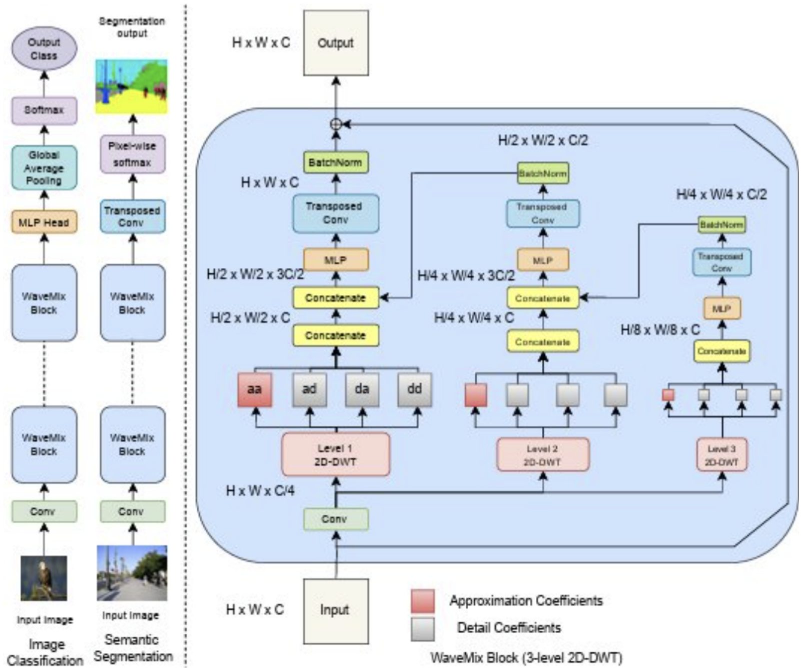

FIGURE 2.4: Wavemix Architecture
source : Wavemix: A resource-efficient neural network for image analysis

uses multi-level 2D Discrete Wavelet Transforms (2D-DWT) to mix spatial information. These transforms decompose input feature maps into multiple frequency bands, enabling multi-scale analysis of visual features. Crucially, WaveMix achieves this with no additional learnable parameters, making it extremely lightweight and well-suited for environments where resource constraints are paramount.

The central component of WaveMix is the WaveBlock, which maintains the same input and output dimensions and applies the wavelet transform to operate on different scales of information. By reducing spatial resolution in a reversible and information-preserving manner, WaveMix allows for efficient representation learning without sacrificing detail. This approach contrasts with traditional convolutions, which typically reduce resolution through pooling or strided operations that can cause information loss.

WaveMix has demonstrated competitive or superior performance across multiple computer vision benchmarks, including ImageNet-1K, Cityscapes, EMNIST, and BSD100. In tasks like semantic segmentation and image super-resolution, it outperforms many traditional and transformer-based methods while using a fraction of the parameters. In the context of generative modeling, WaveMix serves as a promising alternative to resource-heavy attention layers and convolutional blocks. Its integration into diffusion architectures, as we explore in this thesis, shows potential in reducing both training and inference cost while maintaining generative quality.

2.3 Problem Definition

2.3.1 Motivation

The rise of diffusion models has brought unprecedented advances in the field of generative modeling, allowing for the synthesis of highly realistic images across diverse domains. These models offer robust performance and superior image quality, often surpassing traditional generative adversarial networks (GANs) in both diversity and stability. However, their practical deployment is frequently hindered by one key bottleneck: computational inefficiency (Wang, 2024; Croitoru et al., 2023). The denoising process in diffusion models involves repeatedly passing noisy images through large neural networks over several hundred steps, resulting in prohibitively high memory, compute, and time requirements.

At the heart of this inefficiency lies the architecture of the denoising network—most commonly a deep variant of the U-NET. While this architecture is effective in preserving spatial information and reconstructing fine image details, it also introduces an enormous parameter burden. The attention mechanisms used in these networks scale quadratically with input resolution and require extensive memory. Moreover, residual blocks and convolutions further increase the parameter count and computation per step. These architectural choices make diffusion models inaccessible in settings where resources are limited, such as edge devices, real-time applications, or battery-constrained environments.

The growing demand for portable and efficient generative models—spanning fields like mobile photography, augmented reality, and on-device editing—has thus fueled interest in architectural innovations. These innovations aim to retain or improve image quality while reducing model size, inference time, and power consumption. Achieving this balance is a non-trivial challenge, as reductions in capacity often come at the cost of generative fidelity or diversity. This work is motivated by the desire to identify and validate resource-efficient architectural alternatives for diffusion models that overcome these challenges through principled design and empirical evaluation.

2.3.2 Problem Statement

Given the computational limitations and scalability concerns of conventional diffusion architectures, this thesis addresses the following core problem :

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How can we design diffusion model architectures that significantly reduce parameter count and computational cost, while maintaining or improving generative performance across diverse datasets?

This central question encapsulates multiple subproblems:

- How can we optimize resolution changes (downsampling/upsampling) to be more efficient without losing spatial fidelity?
- Can we replace attention mechanisms—which are computationally expensive—with alternatives that retain or improve multi-scale feature extraction?
- Is it possible to move to a latent-space formulation where generation occurs in a lower-dimensional space, thus reducing the workload on the diffusion model itself?
- What architectural components can be re-imagined or removed to improve overall resource efficiency?

Addressing these subproblems requires both theoretical and empirical insight into how generative quality correlates with network structure and complexity. The challenge lies in designing lightweight alternatives that avoid the pitfalls of oversimplification, such as loss of expressivity or poor generalization.

2.3.3 Approach

To systematically address the above problems, we explore several architectural directions, each aimed at a specific aspect of diffusion model efficiency.

2.3.3.1 WaveUnet and its Variants

We introduce WaveUNet, an architecture that replaces standard downsampling and upsampling blocks in U-NET with WaveBlock-inspired modules from the WaveMix family. These blocks use the 2D Discrete Wavelet Transform (DWT) to perform resolution changes in a learnable yet parameter-efficient manner. Variants of WaveUNet also explore non-learnable alternatives (e.g., interpolation-based upsampling) and domain-inspired strategies (e.g., inverse DWT for upsampling).

Each variant is designed to strike a different balance between learnability, computational efficiency, and reconstruction fidelity. These variants allow us to study the trade-offs involved in modifying U-NET's resolution manipulation blocks.

2.3.3.2 Att2Wave

Attention layers, while powerful for capturing long-range dependencies, are notoriously inefficient due to their quadratic complexity with respect to spatial size. In Att2Wave, we remove attention blocks entirely and replace them with wavelet-based blocks that retain multi-scale representational power with near-linear complexity. The core hypothesis is that wavelet transforms can implicitly model hierarchical relationships in images without the cost of explicit attention.

This substitution not only reduces parameter count but also improves scalability to high-resolution inputs. Att2Wave represents a radical departure from transformer-inspired architectures, favoring signal processing principles over direct token interaction modeling.

2.3.3.3 InternImage and DCNs

InternImage brings in Deformable Convolutions (DCNv3) as an alternative to fixed-grid convolutions. By learning spatial offsets and modulations for convolutional kernels, DCNs adapt their receptive fields based on input structure, enabling dynamic feature aggregation with minimal parameter overhead.

We experiment with replacing U-NET residual blocks with InternBlocks, drawing architectural inspiration from vision transformers while preserving the computational efficiency of convolutional layers. The flexibility of InternImage allows for adjustable complexity in different layers, making it suitable for fine-tuning trade-offs between efficiency and expressivity.

2.3.3.4 LiteVAEs

In addition to architectural optimizations in pixel space, we investigate latent-space modeling using LiteVAE (Sadat et al., 2025). The idea is to compress the input image into a compact latent representation via wavelet-based encoders, apply the diffusion process in this lower-dimensional space, and reconstruct the output using a lightweight decoder.

LiteVAE decomposes the input into wavelet sub-bands, applies feature extraction in each band, and aggregates them into a final latent code. This approach significantly reduces the computational burden during sampling, as the diffusion model operates on smaller tensors. By freezing the encoder-decoder weights and training the diffusion model only in the latent space, we also decouple generative modeling from feature learning, leading to faster convergence and more interpretable results.

2.3.4 Expected Outcomes

Through these architectural innovations and controlled experiments, we expect to demonstrate that:

- Resource-efficient architectures such as WaveUNet and Att2Wave can match or outperform traditional U-NETs in FID with far fewer parameters.
- Removing attention via wavelet-based blocks results in comparable or better generative quality while dramatically reducing computation time and memory usage.
- InternImage-inspired blocks provide a middle ground between flexibility and efficiency, offering strong performance with moderate complexity.
- Operating in the latent space using LiteVAE offers compelling trade-offs between speed and generative diversity, especially for real-time or on-device scenarios.

Ultimately, this work aims to lay the foundation for scalable and deployable diffusion models, enabling their use in a wider range of practical applications beyond high-resource cloud environments.

2.3.5 Related Work

The growing literature on generative diffusion models has explored various efficiency-enhancing directions. Ho et al. (2020) introduced DDPMs, laying the foundation for stable, high-fidelity generative modeling. Nichol and Dhariwal (2021) improved sampling speed via DDIM and introduced classifier-free guidance. Recent research has focused on architectural efficiency: WaveMix (Yaqoob et al., 2022) incorporated wavelets into CNNs, while InternImage (Wang et al., 2022) integrated deformable convolutions (Dai et al., 2017; Zhu et al., 2019) into vision backbones. Latent diffusion models (LDMs) proposed by Rombach et al. (2022) compressed inputs before diffusion. Our work expands on these ideas by proposing and benchmarking several novel architecture variants within a unified diffusion framework.

2.4 Architectures and its Variants

In this section, we systematically explore architectural variants aimed at improving the resource efficiency of diffusion models. Each model is designed to address specific computational bottlenecks - whether it's spatial resolution handling, attention overhead, or latent dimensionality. We group these variants by their structural lineage (e.g., U-NET-based, attention-free, latent-space) and evaluate their trade-offs using empirical results in Section 2.5

To reduce the resource demands of diffusion models without compromising generative performance, this work explores several novel and modified architectures. These range from wavelet-inspired modifications of traditional U-NET structures to entirely new attention-free and latent-space-based models. In this section, we provide an in-depth discussion of each architecture, its design motivation, implementation details, and expected benefits.

2.4.1 WaveMixSR

As an early exploration into non-U-NET diffusion architectures, we experimented with using **WaveMixSR**—a wavelet-based model originally designed for super-resolution—as the denoising network in a diffusion pipeline. Unlike other variants that retain U-NET-like hierarchies, this architecture entirely discards encoder-decoder structures, skip connections, and spatial downsampling or upsampling mechanisms.

The WaveMixSR model is composed of repeated `WaveMixSRBlock` modules, which stack `Level1WaveBlocks` and shallow convolutional layers. Temporal information is injected via a small MLP acting on the diffusion timestep, and spatial dimensions remain fixed throughout the forward pass. Each block performs localized wavelet-based spatial mixing, relying solely on frequency-domain interactions for feature extraction.

This approach was motivated by the compactness and simplicity of WaveMix-style models, offering potential gains in memory efficiency and inference speed. However, **experimental results on datasets like CIFAR-10 and MNIST revealed that the absence of hierarchical resolution changes and skip connectivity significantly degraded performance**. Specifically, the model exhibited high FID scores and struggled to capture fine details and long-range dependencies in generated samples.

Although not competitive in its current form, WaveMixSR served as a valuable baseline and proof-of-concept. Its limitations highlighted the importance of U-NET-like architectural priors in diffusion tasks—particularly the benefits of spatial hierarchy and multiscale reconstruction.

2.4.2 UNET-Based Variants

While much of the architectural innovation in this work focuses on replacing U-NET structures entirely or modifying them with wavelet-inspired components, we also explored multiple internal reconfigurations within the standard U-NET design space. These variants aimed to improve parameter efficiency, representation quality, or training stability by modifying core architectural blocks, normalization layers, and the overall network depth.
The following subsections describe these UNET-based modifications in detail.

2.4.2.1 Residual-Replaced U-NET

This variant is based on the original U-NET architecture but replaces the conventional residual blocks with a custom `ResidualWaveBlock` designed to enhance multiscale representation while maintaining parameter efficiency. These blocks are inspired by frequency-domain processing and introduce wavelet-inspired decompositions directly within the residual path.

The standard residual block in U-NET consists of two convolutional layers separated by normalization and activation, along with a projection of the time-step embedding and an identity or 1×1 convolution shortcut connection. In contrast, the `ResidualWaveBlock` significantly alters the internal structure. It starts by compressing the input through a 1×1 convolution followed by a wavelet-based decomposition. This produces a low-frequency approximation and high-frequency components, which are concatenated to form the input to a lightweight feedforward module.

The feedforward path consists of:

- A 1×1 expansion layer followed by a GELU activation and dropout.
- A subsequent dimensionality reduction through another 1×1 convolution.
- A 4×4 transposed convolution to re-expand spatial resolution, capped with a batch normalization layer.

Additionally, the time embedding is projected and added post-decomposition, influencing the concatenated wavelet representation. If the input and output channels differ, a learnable 1×1 convolution is used as the skip connection; otherwise, identity mapping is used.

This design introduces a form of **frequency-aware feature routing**, where the residual block does not just operate in the spatial domain but explicitly manipulates and reweighs multi-frequency information. The added upsampling at the end of the block acts as a resolution-boosting step within the residual path. This is a notable departure from conventional residual blocks that typically preserve resolution throughout.

While conceptually interesting and parameter-efficient, empirical results showed that this variant did not lead to significant improvements over the baseline U-NET. On datasets like CIFAR-10 and MNIST, the model's FID was slightly higher than the vanilla version. This suggests that although frequency decomposition offers theoretical benefits, its integration into residual blocks may need further tuning or a more suitable pairing with spatial-resolution handling layers to be fully effective in diffusion models.

2.4.2.2 Deep Residual U-NET (4-Stage)

This variant builds directly on the Residual-Replaced U-NET variant discussed in the previous section but increases the architectural depth from the typical 3-level U-NET to a 4-level hierarchy. This deeper version incorporates additional downsampling and upsampling stages, leading to a larger receptive field and more abstract feature representation.

The aim of this variant was to evaluate whether a deeper U-NET with updated residual blocks could yield performance improvements over the shallower baseline. The deeper structure enables the network to learn more complex representations across a wider range of spatial scales. Each resolution level is connected by skip connections, and residual blocks at every level use the modified design described in Residual-Replaced U-NET.

Despite its increased capacity, the model did not yield improvements in FID and suffered from longer training times. The lack of gains suggests that simply deepening the U-NET does not necessarily translate to better generative quality, especially in diffusion contexts where model expressivity must be balanced with noise robustness.

2.4.2.3 Normalized U-NET with Revised Layer Ordering

This introduces a series of internal adjustments to the standard U-NET architecture with the goal of improving **training stability and feature propagation**. The changes include:

- Replacing **GroupNorm** with **BatchNorm** in residual blocks.
- Reordering layers such that normalization follows the convolution and activation (Conv → Activation → Norm), inspired by empirical stability findings.
- Replacing GroupNorm with **RMSNorm** (Zhang et al., 2022) in attention blocks to reduce variance explosion in deep layers.

These design choices were motivated by recent works such as Zhang et al. (2022) (RMSNorm) and Liu et al. (2020) (normalization layer orderings), which argue that the placement and type of normalization have a significant effect on gradient flow and convergence in deep neural networks. The updated normalization scheme in this variant sought to reduce training instability and improve generalization across datasets.

However, despite these theoretically sound changes, empirical results indicated only marginal improvements, if any, in FID, and no consistent trend across datasets. The findings suggest that while normalization tweaks may contribute to smoother training dynamics, they are not sufficient in isolation to improve the final generative performance of diffusion models.

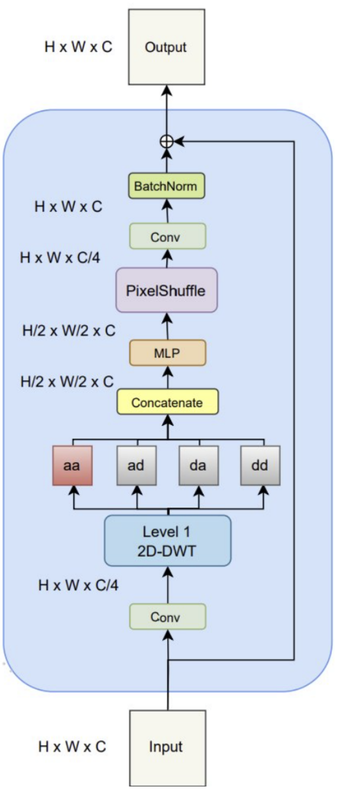

FIGURE 2.5: Waveblock Architecture

2.4.3 WaveUNet

2.4.3.1 Motivation

The U-NET architecture has long served as the backbone of diffusion models due to its ability to preserve spatial resolution through skip connections and its robustness in image reconstruction tasks. However, its standard downsampling and upsampling blocks are computationally expensive and contribute significantly to the model's parameter count and memory consumption. In an effort to make these operations more efficient, we propose WaveUNet, which replaces the conventional resolution manipulation layers in U-NET with wavelet-based blocks drawn from the WaveMix family.

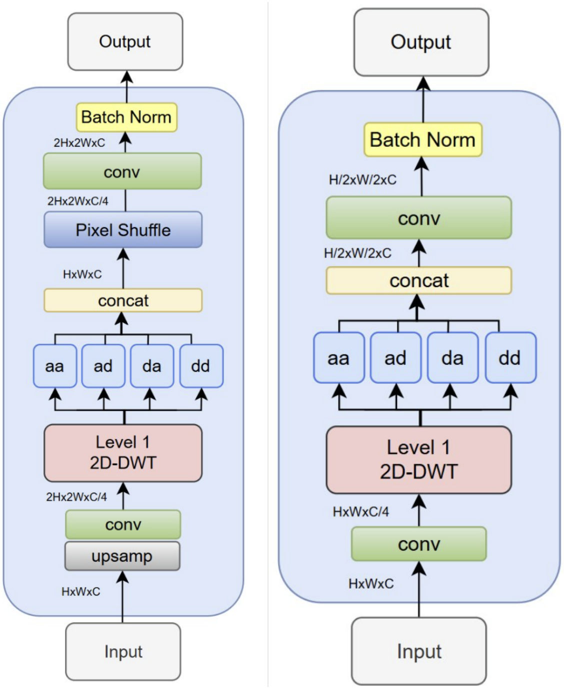

FIGURE 2.6: WaveUnet Upsampling(left) and Downsampling(right) blocks

2.4.3.2 Architecture Overview

WaveUNet modifies the encoder-decoder structure of the U-NET by substituting:

- Downsampling blocks with a version of the WaveBlock that excludes convolution layers but performs a Discrete Wavelet Transform (DWT) to reduce spatial resolution.
- Upsampling blocks with PixelShuffle-based mechanisms inspired by WaveMixSR-v2 (Jeevan et al., 2024b), which facilitate resolution restoration using sub-pixel convolutions instead of transposed convolutions or interpolation.

This design introduces a form of learnable resolution change that retains more signal information than conventional pooling and offers finer control over feature reassembly. Importantly, the use of DWT introduces minimal additional parameters, preserving model compactness.

2.4.3.3 Performance Considerations

By integrating wavelet transforms, WaveUNet introduces multi-scale feature representation inherently into the architecture. This helps capture fine and coarse details simultaneously, enhancing generative fidelity. The architecture was empirically observed to produce lower FID scores on datasets like CIFAR-10 while using significantly fewer parameters compared to the standard U-NET. Moreover, GPU memory usage and inference time were also favorably impacted, making WaveUNet a strong candidate for practical deployment.

2.4.4 Variants of WaveUnet

To better understand the trade-offs in architectural design, we explore multiple variants of WaveUNet, each making different compromises between efficiency and expressivity.

2.4.4.1 Hybrid WaveUNet-I

This variant retains the wavelet-based downsampling from WaveUNet but reverts to convolution + interpolation for upsampling, mimicking the original U-NET's decoder structure. The goal is to determine whether maintaining a familiar, parameter-rich decoder can improve reconstruction without a significant increase in overall complexity.

2.4.4.2 Hybrid WaveUNet-II

In this design, we combine WaveBlock-based downsampling with purely non-parametric interpolation for upsampling. This pushes the efficiency frontier further by avoiding any learnable operations in the decoder. The trade-off is a potential degradation in image quality, but it serves as an important baseline for understanding the minimal requirements for acceptable generative performance.

2.4.4.3 Minimalist U-NET (Pooling + Interpolation)

This is the most minimalistic variant, using average pooling for downsampling and bilinear interpolation for upsampling. It eliminates all learnable resolution-changing operations, serving as a lower bound for model capacity. Surprisingly, this simple architecture still yields reasonable FID scores, reinforcing the idea that much of the performance in diffusion models comes from the denoising process rather than the complexity of the network itself.

2.4.4.4 Wavelet-Upsampled WaveUNet

Here, we experiment with inverse discrete wavelet transforms (IDWT) in the decoder to reconstruct spatial resolution. By maintaining a wavelet framework in both downsampling and upsampling, this variant ensures consistency and leverages the invertibility of DWT/IDWT pairs. It is particularly promising for scenarios requiring high-frequency detail preservation.

2.4.4.5 Residual-Replaced WaveUnet

This variant builds on the standard UNET architecture but replaces the conventional residual blocks with wavelet-based residual blocks for better parameter efficiency and multiscale feature extraction. The resolution manipulation—downsampling and upsampling—is also handled by the WaveUNet-style blocks, using discrete wavelet transforms and PixelShuffle operations respectively.

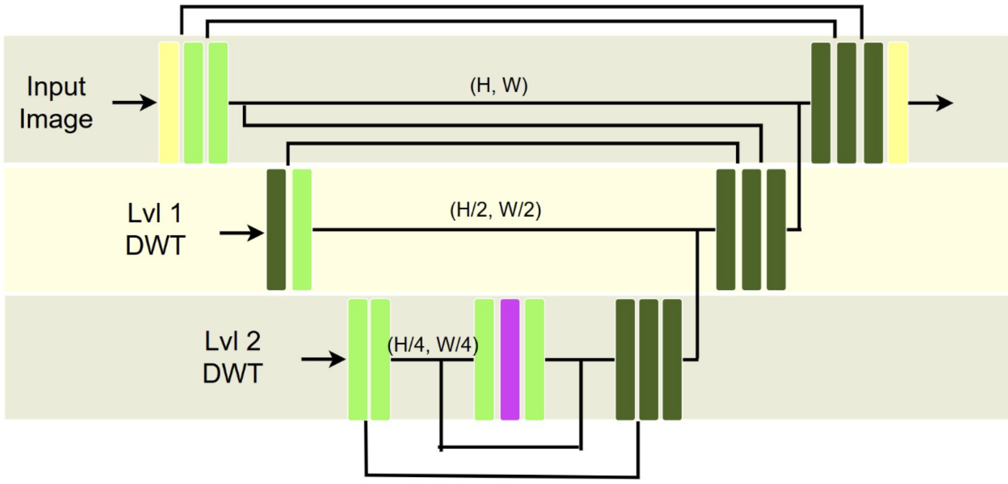

FIGURE 2.7: MultiPath U-NET Architecture
Contains Conv(yellow), Residual(green) and Attention(magenta) blocks

Residual-Replaced WaveUnet is designed to capture the advantages of both worlds: the robust feature learning of the original UNET backbone and the efficiency of wavelet-based processing. The wavelet residual blocks operate across frequency bands, allowing them to capture both global structure and local textures efficiently. This hybrid architecture aims to balance expressivity and efficiency, offering strong reconstruction performance at a significantly reduced computational cost.

2.4.5 MultiPath UNet

Inspired by multi-resolution analysis, the MultiPath UNet expands the idea of wavelet integration by explicitly processing multiple frequency bands. Instead of a single forward pass, the input image is decomposed into three DWT channels (LL, LH, HL, HH), each capturing different spatial-frequency characteristics. These components are then passed through separate encoding paths within the U-NET architecture.

This design allows for parallel feature learning across different granularity levels, encouraging specialization and more expressive latent representations. While computationally heavier than the other variants, MultiPath UNet has shown potential for improving FID at higher resolutions or in datasets with complex textures and patterns.

2.4.5.1 MultiPath U-NET with Bilinear Upsampling

This baseline variant uses a **2-level Discrete Wavelet Transform (DWT)** to decompose the input into four frequency components: LL, LH, HL, and HH. Each component is processed by a distinct encoding path within the UNet, allowing the model to specialize in features at different spatial scales. These paths are then merged at the bottleneck and passed through a common decoder.

For upsampling, **bilinear interpolation** is used in the decoder to restore spatial resolution. The core blocks used are standard convolutional and residual blocks, with attention modules preserved from the original U-NET design. This variant emphasizes multiscale feature learning through explicit frequency band specialization, and aims to test whether parallel wavelet streams can outperform standard single-path designs.

2.4.5.2 MultiPath U-NET with Bicubic Upsampling

This variant is identical in structure to the structure used in the previous section but replaces bilinear interpolation in the decoder with **bicubic interpolation**. The motivation behind this substitution is to evaluate whether the smoother gradients and more expressive interpolation dynamics of bicubic upsampling lead to improved generative performance or image sharpness.

No additional modifications are made to the encoder, attention layers, or residual blocks. Empirical evaluations help quantify whether the interpolation method in itself plays a critical role in generation fidelity when combined with multiscale decomposition.

2.4.5.3 MultiPath Att2Wave

In this variant, we retain the same multi-frequency, multi-path architecture but **replace all attention blocks with** `Lvl1WaveBlocks`, mirroring the substitution strategy used in the Att2Wave variants developed in section 2.4.6. The motivation here is twofold - to reduce the parameter overhead introduced by self-attention, and to test whether frequency-based spatial mixing via WaveBlocks is synergistic with the wavelet decomposition already present in the encoder.

This variant combines the benefits of **frequency-aware resolution decomposition** (via DWT) and **attention-free spatial interaction** (via WaveBlocks). It provides insight into whether multiscale wavelet modeling at both macro (decomposition) and micro (block-level) levels can fully replace the need for explicit attention mechanisms in high-resolution generative modeling.

2.4.6 Att2Wave: Removing Attention with WaveBlocks

2.4.6.1 Motivation

Attention mechanisms, especially those used in transformers, offer excellent modeling capabilities for long-range dependencies but come at a high computational cost. Their quadratic complexity with respect to input size becomes a bottleneck in high-resolution image synthesis. As an alternative, we propose Att2Wave, a model that replaces all attention layers with WaveBlocks, offering a more efficient yet expressive mechanism for spatial feature interaction.

2.4.6.2 WaveBlocks as Attention Alternatives

WaveBlocks operate by decomposing feature maps via 2D wavelet transforms and reassembling them without requiring any attention weights or matrix multiplications. Their near-linear complexity (O(n log n)) makes them highly suitable for replacing attention layers in resource-constrained settings. They also introduce multi-scale feature extraction inherently, without explicit token-to-token interactions, which helps preserve global context.

2.4.6.3 Performance and Trade-offs

While removing attention may seem like a loss in modeling power, empirical results suggest that Att2Wave maintains, and sometimes improves, FID scores compared to standard diffusion models. The parameter efficiency is significant, and the memory footprint is much smaller. The only trade-off is in interpretability—wavelet-based operations do not provide explicit attention maps, making it harder to analyze which parts of the image the model focuses on.

2.4.7 Variants of Att2Wave

While the base Att2Wave architecture offers significant efficiency improvements by replacing attention mechanisms with WaveBlocks, several variants were explored to study the trade-offs between depth, feedforward capacity, and architectural modifications. These variants help assess how wavelet-based spatial mixing scales with model capacity and network depth, and whether additional wavelet-based modifications can further enhance generative performance.

The following subsections describe each variant in detail.

2.4.7.1 Att2Wave

This is the baseline version of the Att2Wave architecture, where each attention block from the original U-NET-based diffusion model is replaced by a lightweight `Lvl1WaveBlock`. This design ensures that each wavelet-based block introduces only minimal computational overhead while providing effective multi-scale spatial mixing.

`Lvl1WaveBlock` is the core spatial mixing unit in this architecture. It is parameterized by two tunable hyperparameters:

- `mult` – a multiplier for internal channel expansion. It controls the width of the intermediate representation relative to the input channels.
- `ff` – the dimensionality of the feedforward layer that follows wavelet decomposition. It controls the expressive capacity of each block.

Unless otherwise stated, we use `mult = 1` and `ff = 16` across most variants. These values were chosen as they strike a balance between efficiency and representational power, particularly in low-to-mid-resolution settings.

The primary goal of this configuration is to verify whether such a minimal substitution is sufficient to replace attention altogether. The results indicate that even with this small configuration, the model performs on par with attention-based models on datasets like CIFAR-10, making it a strong candidate for highly resource-efficient generative tasks.

2.4.7.2 Att2Wave–High Capacity (ff = 64)

This variant is structurally identical to Att2Wave but increases the feedforward expansion dimension to `ff=64`. By expanding the intermediate representation within each WaveBlock, this high-capacity variant introduces more expressive capacity to capture complex spatial features while still avoiding the computational complexity of attention mechanisms.

This variant serves to explore whether deeper intra-block transformations yield improved image fidelity, especially on more detailed datasets like CelebA and Tiny ImageNet. It retains all the benefits of wavelet-based spatial operations while offering a modest increase in model capacity.

2.4.7.3 Att2Wave–Deep Hierarchy (4-Level U-NET)

This variant builds on the original Att2Wave structure but increases the **depth of the network to four layers**, creating a deeper U-NET-style hierarchy. Each resolution level is equipped with WaveBlocks, maintaining full replacement of attention modules with wavelet-based alternatives.

This architecture is intended to evaluate how deep, purely wavelet-driven models scale in terms of generative quality. The added depth allows the model to capture more abstract features while keeping the parameter count lower than attention-rich counterparts. Att2Wave–Deep Hierarchy strikes a balance between model expressivity and efficiency.

2.4.7.4 Att2Wave–Fully Wavelet-Based

This is an enhanced version of Att2Wave–Deep Hierarchy that also replaces standard residual blocks with **wavelet-based residual blocks**, inspired by the design used in Residual-Replaced WaveUnet. This results in a fully wavelet-based architecture across all dimensions—resolution changes, spatial mixing, and residual feature learning.

By leveraging both the hierarchical depth of Att2Wave–Deep Hierarchy and the parameter efficiency of wavelet residuals, this variant maximizes multi-scale representational capacity while remaining lightweight. Experimental results showed that Att2Wave–Fully Wavelet-Based achieved better high-frequency detail recovery and improved FID scores in challenging domains, making it a promising architecture for resource-constrained high-resolution image synthesis.

2.4.7.5 Att2Wave–Stacked Blocks

This architecture modifies the original the baseline version of Att2Wave architecture by introducing **two sequential** `Lvl1WaveBlocks` in the middle path at each resolution level, instead of just one. This modification increases the representational depth of each spatial mixing step while keeping the overall architecture and parameter count relatively lightweight.

The motivation behind this design is to test whether stacking multiple lightweight WaveBlocks provides additional modeling power, particularly in capturing more complex local and mid-range interactions. By allowing a deeper flow within each spatial resolution, this variant is able to better refine activations before passing them through skip connections or resolution changes.

Empirical results showed that this variant yielded modest improvements in FID over the base Att2Wave model, especially on datasets with richer textures or more structured objects. The trade-off was a slight increase in runtime and parameters, but the simplicity and regularity of the double-WaveBlock pattern make it an attractive extension for improved expressivity with minimal architectural change.

2.4.7.6 Att2Wave–PixelShuffle Upsampling

Att2Wave–PixelShuffle Upsampling retains the overall structure of the original Att2Wave architecture but replaces the WaveBlocks with a **PixelShuffle-based variant** inspired by WaveMixSRv2 (Jeevan et al., 2024b). Instead of using interpolation-based or transposed convolution for upsampling, this variant leverages the sub-pixel rearrangement mechanism of PixelShuffle to perform learnable, efficient upsampling within the wavelet-based block.

The integration of PixelShuffle allows the network to perform spatial resolution changes with minimal artifacts and enhanced sharpness in the output. This technique has been effective in super-resolution tasks and is adapted here to enhance detail fidelity in the denoising steps of the diffusion process.

In experiments, this variant maintained competitive FID scores while offering improvements in visual sharpness and fine detail reconstruction. This variant represents a hybrid between classical wavelet-based spatial mixing and modern efficient upsampling strategies, and is particularly effective for tasks where output crispness and texture realism are prioritized.

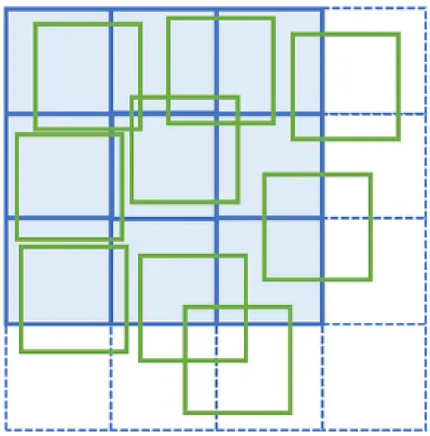

FIGURE 2.8: Deformable Convolution
source : medium.com/@alejandro.itoaramendia/deformable-convolutional-networks-a-complete-guide-eadd9f1f8ce2

2.4.8 InternImage with Deformable Convolutions (DCNv3)

2.4.8.1 Overview

InternImage (Wang et al., 2023) integrates DCNv3 blocks - an advanced form of deformable convolutions (see section 2.4.8.2), into vision architectures. These convolutions dynamically learn spatial offsets, allowing the kernel to adapt to object boundaries and structural variations in the input. This flexibility enables DCNs to capture both local detail and long-range context with fewer parameters than traditional convolution or attention layers.

2.4.8.2 Deformable Convolution

Deformable convolutions (Dai et al., 2017; Zhu et al., 2019) enhances standard convolution by introducing learnable offsets to the sampling grid, allowing the convolutional kernel to adaptively focus on relevant regions of the input feature map. Unlike traditional convolution, which uses a fixed grid, deformable convolution adjusts the sampling locations dynamically based on the input content. For an input feature map x and output feature map y, the deformable convolution operation at position p_0 is defined as:

$$y(p_0) = \sum_{k=1}^{K} w_k \cdot x(p_0 + p_k + \Delta p_k),$$

where w_k represents the weights of the convolutional kernel, p_k denotes the predefined offset for the k-th sampling location in the regular grid (e.g., a 3x3 grid), and Δp_k is the learnable offset predicted by an additional convolutional layer. These offsets Δp_k are typically learned from the input feature map via a separate convolution, enabling the model to capture more flexible and context-aware patterns, improving performance in tasks like object detection and image segmentation.

2.4.8.3 Benefits of Deformable Convolution

Deformable Convolutional Networks (DCNs) offer significant advantages over standard convolution, global attention, and local attention by combining adaptive spatial aggregation, effective long-range dependency modeling, and high computational efficiency. DCNs dynamically adjust sampling locations through learned offsets, enabling content-aware feature extraction that excels in handling geometric variations in tasks like object detection and segmentation. Unlike standard convolution's fixed grids or local attention's limited windows, DCNs capture distant relationships efficiently, while avoiding the quadratic complexity of global attention. This makes DCNs a powerful, memory-efficient solution for vision tasks requiring flexibility and performance.

2.4.8.4 Intern Block

The Intern block, a core component of the InternImage architecture, is designed to leverage deformable convolution (DCNv3) as its fundamental operator, combining the strengths of convolutional neural networks (CNNs) with adaptive spatial aggregation. Each Intern block integrates a deformable convolution layer, layer normalization (LN), and a feed-forward network (FFN), structured similarly to Transformer blocks but using DCNv3 to dynamically adjust the sampling grid based on input content. The operation can be seen in Fig. 2.10 where DCNv3 denotes the deformable convolution operation, LN is layer normalization, FFN is a feed-forward network, and γ_1, γ_2 are learnable scaling parameters for layer scaling. This structure enables InternImage to capture large effective receptive fields and adaptively aggregate spatial features, enhancing performance in tasks like object detection and segmentation, as demonstrated by its record-breaking 65.4 mAP on COCO test-dev.

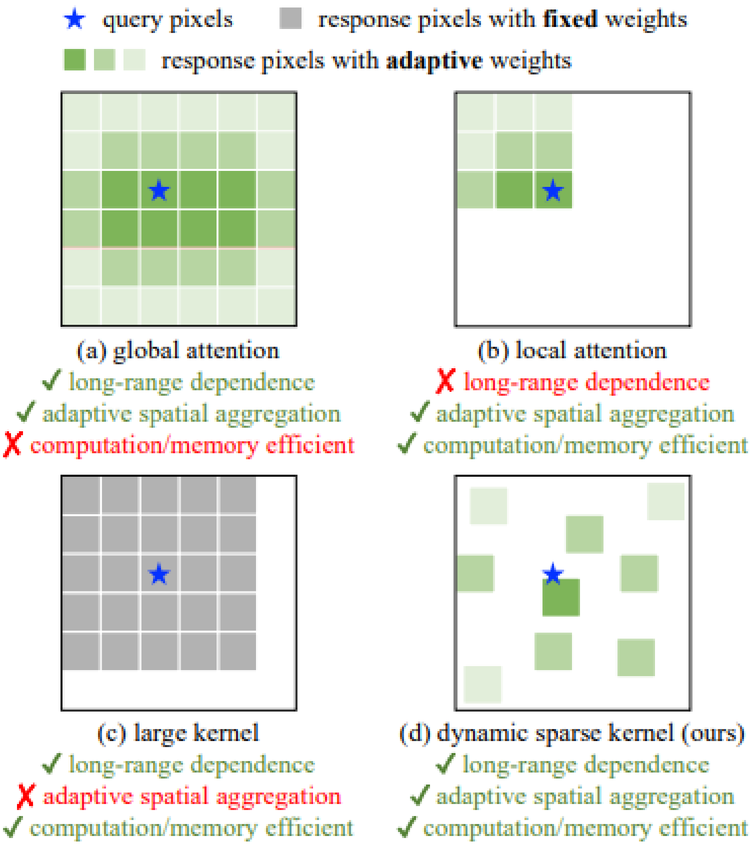

FIGURE 2.9: Comparisons of different core operators
source : InternImage: Exploring Large-Scale Vision Foundation Models with Deformable Convolutions

2.4.8.5 Integration into Diffusion Framework

In our work, we use a U-NET's like arhitecture replacing residual blocks with Intern-Blocks. Downsampling is performed using strided convolutions, while upsampling uses either transposed convolutions or sub-pixel shuffling. This structure mimics transformer pipelines while preserving the efficiency of CNNs.

Other variants using Intern Blocks by making slight modifications in the architectures like changing the sampling methods used, initial STEP layer, and final CNN layer which did not produce results anu better than the finalised architecture.

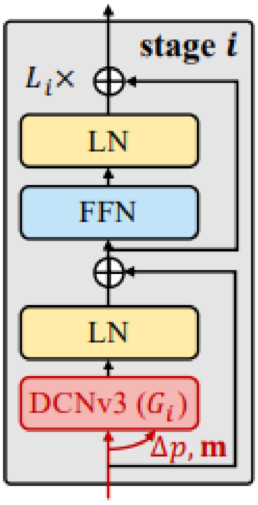

FIGURE 2.10: Intern Block Architecture
source : InternImage: Exploring Large-Scale Vision Foundation Models with Deformable Convolutions

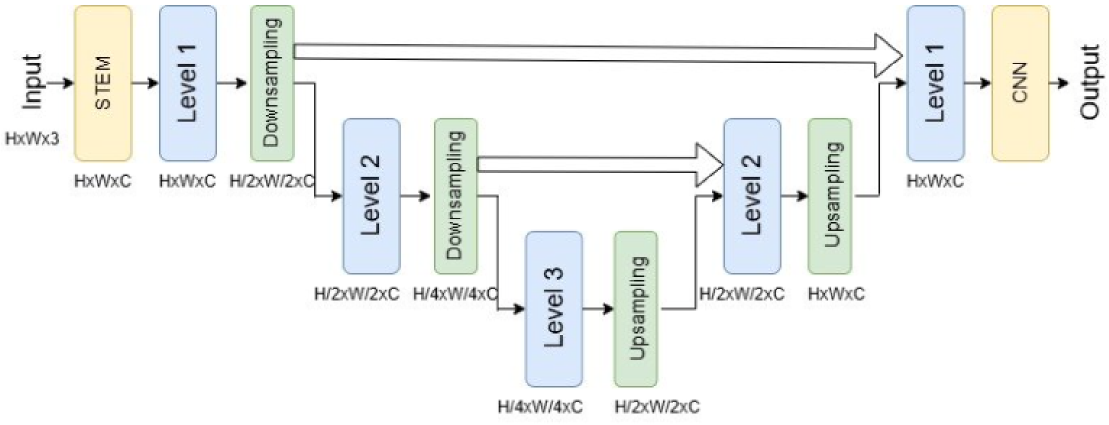

FIGURE 2.11: Intern U-Net Architecture

2.4.8.6 Observations

InternImage-based diffusion models demonstrate competitive performance in terms of FID, particularly when used in combination with skip connections. However, their gains are less pronounced compared to the wavelet-based architectures in terms of parameter reduction. They are best viewed as a middle-ground solution—offering better generalization than minimalistic models and better efficiency than full U-NETs with attention.

2.4.9 LiteVAE for Latent Diffusion

2.4.9.1 Background

Variational Autoencoders or VAEs (Kingma and Welling, 2022) are a powerful class of generative models that combine principles from deep learning and Bayesian inference to create a probabilistic framework for encoding and generating data, particularly images. Unlike traditional autoencoders, which focus solely on deterministic data compression and reconstruction, VAEs introduce a structured approach to modeling data distributions. They achieve this by learning a compact, continuous latent representation of input data, which can then be used for tasks such as image synthesis, denoising, data augmentation, and anomaly detection. The probabilistic nature of VAEs allows them to capture the underlying variability in data, making them particularly effective for generative tasks where the goal is to produce diverse yet realistic outputs.

At the core of the VAE framework is the idea of encoding high-dimensional input data, such as images, into a lower-dimensional latent space. This latent space serves as a compressed representation of the input, capturing its essential features while reducing computational complexity. By sampling from this latent space, VAEs can generate new data points that resemble the original dataset, making them valuable for applications like image generation, style transfer, and data interpolation. The training of VAEs involves optimizing two objectives: a reconstruction loss, which ensures that the generated output closely matches the input, and a regularization term, often based on the Kullback-Leibler (KL) divergence, which encourages the latent space to follow a specific distribution, typically a Gaussian. This dual objective enables VAEs to balance fidelity to the input data with the ability to generate diverse samples.

The appeal of VAEs lies in their ability to reduce the dimensionality of complex datasets, enabling the use of smaller, more efficient neural networks for generative tasks. By projecting high-dimensional inputs into a lower-dimensional latent space, VAEs reduce the computational and memory requirements of downstream tasks, making them suitable for resource-constrained environments. Furthermore, the probabilistic framework of VAEs allows for uncertainty quantification, which is critical in applications such as medical imaging, where understanding the confidence of generated outputs can be as important as the outputs themselves. This flexibility has made VAEs a cornerstone in fields ranging from computer vision to natural language processing.

LiteVAE, a recently proposed advancement in the VAE family, builds upon this foundation by incorporating wavelet-based decomposition and a parameter-efficient design. Traditional VAEs often rely on convolutional neural networks (CNNs) to process images, which can be computationally intensive and parameter-heavy, especially for high-resolution inputs. LiteVAE addresses these challenges by leveraging wavelet transforms, a mathematical tool that decomposes images into frequency components, capturing both spatial and frequency information efficiently. This decomposition allows LiteVAE to represent complex image structures with fewer parameters, reducing the model's memory footprint and computational demands while maintaining high-quality reconstruction and generation capabilities.

The parameter-efficient design of LiteVAE makes it particularly well-suited for deployment on resource-constrained devices, such as mobile phones or embedded systems, where computational power and memory are limited. By combining the probabilistic strengths of VAEs with the efficiency of wavelet-based methods, LiteVAE offers a compelling solution for real-world applications that require fast, lightweight,

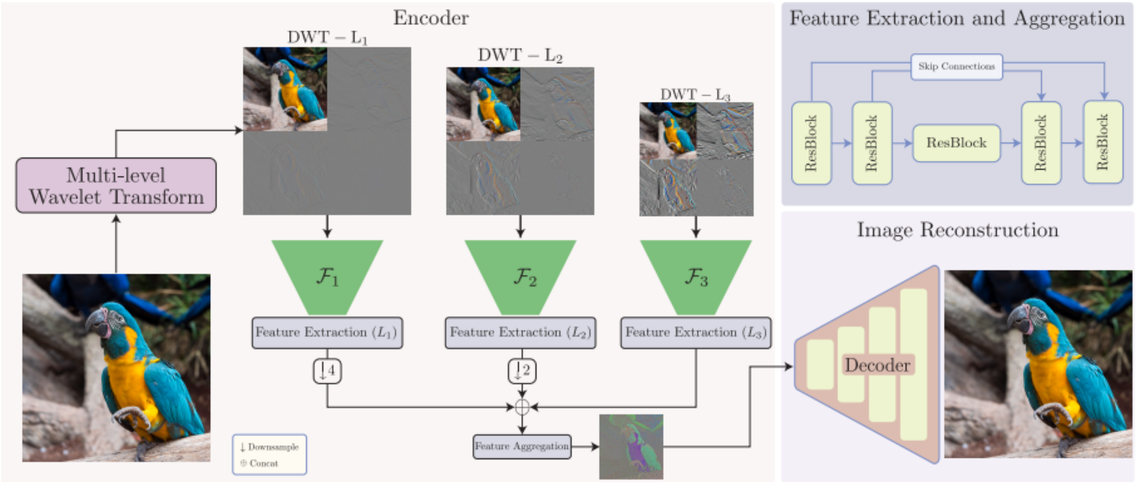

FIGURE 2.12: An overview of LiteVAE
source : LiteVAE: Lightweight and Efficient Variational Autoencoders for Latent Diffusion Models

and scalable generative models. This architecture retains the generative flexibility of traditional VAEs while introducing innovations that enhance its practicality for modern machine learning tasks. As a result, LiteVAE represents a significant step forward in making generative modeling more accessible and efficient without sacrificing performance.

2.4.9.2 Design and Workflow

LiteVAE (Sadat et al., 2025) is a wavelet-based autoencoder that achieves the reconstruction quality of standard VAEs with significantly lower complexity. The method consists of three main components (see Fig 2.12):

1. **Wavelet Processing:** Each image x is first processed via a multi-level Discrete Wavelet Transform (DWT) to obtain the corresponding wavelet coefficients $\{x^l_L, x^l_H, x^l_V, x^l_D\}$ at level l. To achieve an $8\times$ downsampling, we use three wavelet levels (i.e., $l \in \{1, 2, 3\}$). These features extract multiscale information from x.

2. **Feature Extraction and Aggregation:** The wavelet coefficients $\{x_L^l, x_H^l, x_V^l, x_D^l\}$ are separately processed via a feature-extraction module F_l to compute a multiscale set of feature maps:

$$F_l(\{x_L^l, x_H^l, x_V^l, x_D^l\})$$

   The features are then combined using a feature-aggregation module F_{agg} that takes the output of each F_l and computes the latent code z. We use a UNet-based architecture (similar to ADM **?**) without spatial down/upsampling layers for both feature extraction and aggregation (see Appendix B for discussion).

3. **Image Reconstruction:** Finally, a decoder network D processes the latent code z and computes the reconstructed image $\hat{x} = D(z)$. We use the same decoder network as in SD-VAE for D.

2.4.9.3 Training LiteVAE

LiteVAE employs multiple loss functions to optimize reconstruction quality, latent space regularity, and training stability.

1. **Reconstruction Loss (L1 Loss)**

   - **Purpose:** Measures pixel-level differences between the input image x and the reconstructed image $\tilde{x} = \mathcal{D}(\mathcal{E}(x))$.
   - **Form:**

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_x \left[\|x - \tilde{x}\|_1 \right]$$

   - **Role:** Encourages accurate pixel-level reconstruction. L1 loss is preferred over L2 to reduce blurring and preserve sharp details.

2. Perceptual Loss

- **Purpose:** Captures high-level semantic similarities between x and $\tilde{x}$.
- **Form:**

$$\mathcal{L}_{\text{perc}} = \mathbb{E}_x \sum_i \|\phi_i(x) - \phi_i(\tilde{x})\|_2^2$$

  where ϕ_i denotes features from a pretrained network (e.g., VGG-16).
- **Role:** Helps ensure reconstructions lie on the natural image manifold, improving perceptual quality.

3. Kullback-Leibler (KL) Divergence Loss

- **Purpose:** Regularizes the latent space to follow a standard normal distribution $\mathcal{N}(0, I)$.
- **Form:**

$$\mathcal{L}_{\text{KL}} = \mathbb{E}_x \left[D_{\text{KL}} \left(q(z|x) \| \mathcal{N}(0, I) \right) \right]$$

  where $q(z|x) = \mathcal{N}(\mu(x), \sigma(x))$ is the approximate posterior.
- **Role:** Encourages well-structured and disentangled latent variables for sampling and generative tasks.

4. Adversarial Loss

- **Purpose:** Improves realism of the reconstructed image using a discriminator.
- **Form:**

$$\mathcal{L}_{\text{adv,G}} = \mathbb{E}_x \left[\log(1 - D(\tilde{x})) \right] \quad \text{(Generator)}$$

$$\mathcal{L}_{\text{adv,D}} = \mathbb{E}_x \left[\log D(x) \right] + \mathbb{E}_x \left[\log(1 - D(\tilde{x})) \right] \quad \text{(Discriminator)}$$

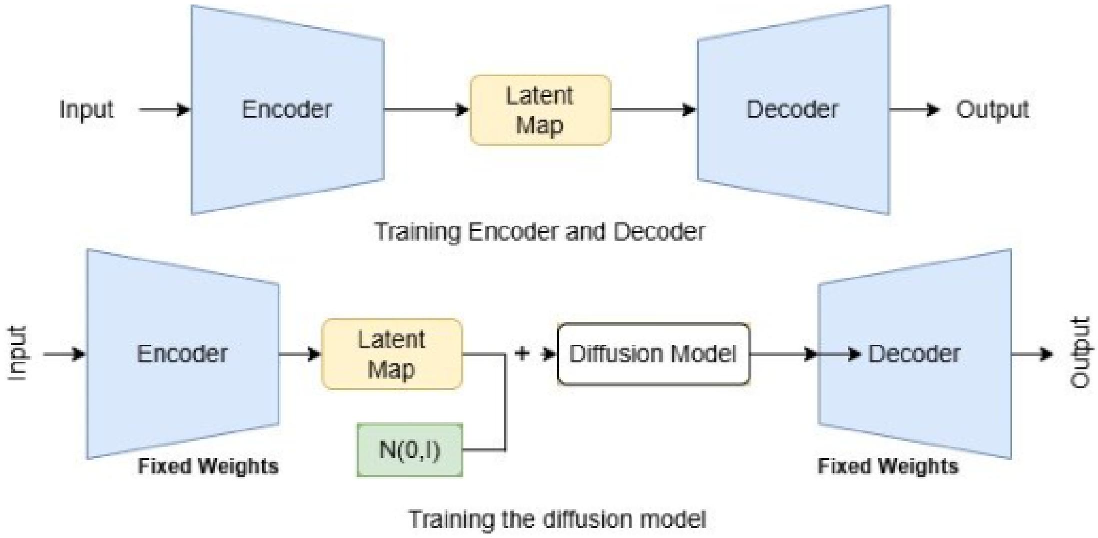

FIGURE 2.13: An overview of LiteVAE training

- **Implementation:** Uses a UNet-based discriminator for pixel-wise feedback. Loss weight is constant for stability in mixed-precision settings.
- **Role:** Sharpens reconstructions and reduces artifacts by aligning with real image distributions.

5. Wavelet-Specific Loss

- **Purpose:** Ensures frequency-band fidelity between original and reconstructed images.
- **Form:**

$$\mathcal{L}_{\text{wavelet}} = \sum_{b \in \{\text{LL, LH, HL, HH}\}} \mathbb{E}_x \left[\|\text{DWT}_b(x) - \text{DWT}_b(\tilde{x})\|_1 \right]$$

where DWT_b extracts the b-th wavelet band.

- **Role:** Preserves low- and high-frequency detail in reconstructions, aligning with LiteVAE's wavelet-based design.

2.4.9.4 Training Diffusion Models Using LiteVAE Latent Map

Once LiteVAE is trained and capable of encoding images into a compact latent space, a diffusion model is trained directly in this latent space to model the distribution of encoded representations. Specifically, for each training image x, the pretrained LiteVAE encoder $\mathcal{E}$ produces a latent code $z = \mathcal{E}(x)$. The diffusion model is then trained to learn the distribution $p(z)$ using a denoising score matching objective. At each diffusion timestep t, Gaussian noise is progressively added to z to produce z_t, and the model learns to reverse this corruption process by predicting either the noise ϵ or the original latent z conditioned on z_t and t. The training objective typically takes the form:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{z,\epsilon,t} \left[\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2 \right],$$

where $\epsilon \sim \mathcal{N}(0, I)$ and ϵ_θ is the noise prediction network. By operating in LiteVAE's wavelet-informed latent space, the diffusion model benefits from more compact, structured representations, leading to faster convergence, reduced computational costs, and higher fidelity generation compared to training directly in pixel space.

2.4.9.5 Benefits and Limitations

This approach allows the diffusion model to operate in a low-dimensional space, dramatically reducing the number of floating-point operations per sampling step. Inference is faster, and the model footprint is smaller. However, the trade-off lies in the reliance on the VAE's capacity to preserve visual fidelity, and the complexity of pretraining the encoder-decoder pair. Nonetheless, LiteVAE remains a promising direction for scalable generative modeling.

2.5 Experimentation and Results

A rigorous empirical evaluation was conducted to assess the effectiveness of the proposed resource-efficient architectures. This section details the experimental setups, datasets used, model configurations, evaluation metrics, and the results obtained across multiple baselines and variants. The primary metric used for evaluation is Fréchet Inception Distance (FID), which measures the quality and diversity of generated images. Additionally, we compare parameter counts to assess the resource efficiency of each architecture.

2.5.1 Datasets

To validate the generalizability and robustness of our architectural variants, experiments were conducted across a diverse set of benchmark datasets:

- MNIST: A grayscale handwritten digit dataset of 28×28 images, used primarily to observe the performance of lightweight models on simpler visual domains.
- CIFAR-10: A standard benchmark for image generation, consisting of 60,000 32×32 color images across 10 classes.
- Tiny ImageNet: A subset of ImageNet with 200 classes, where each image is resized to 64×64 pixels, offering increased diversity and complexity over CIFAR-10.
- CelebA: A dataset of over 200,000 celebrity face images, used to evaluate the models on structured, high-resolution, and identity-preserving tasks.

These datasets collectively cover a wide range of visual characteristics and complexity, allowing for a holistic evaluation of the proposed models.

2.5.2 Evaluation Metrics

The primary evaluation metric used is the Fréchet Inception Distance (FID). FID scores were computed using a standard Inception-v3 model and compared across

1. Original U-NET-based diffusion model (baseline)
2. WaveUNet and its variants
3. Att2Wave
4. InternImage-based models
5. LiteVAE + diffusion

For fairness, the same number of diffusion timesteps, loss functions, and sampling procedures were used across all models.
In addition to FID, we measure the number of trainable parameters, an indicator of memory and storage footprint.

2.5.3 Results on CIFAR-10

We evaluated a wide range of architectural variants on CIFAR-10 to understand the trade-offs between model complexity and generative performance. The results are shown in Tables 2.1 and 2.2. Across both tables, FID is used as the primary performance metric.

On CIFAR-10, we observed several clear trends in model performance and efficiency:

TABLE 2.1: Experimental Results on CIFAR-10: Across Model Variants

2gray!5white

gray!25 **Model**	**Parameters**	white**FID**
green!20UNET	13.800003 M	white7.29
red!25WavemixSR	12.992844 M	white50.046
yellow!25WaveUNET	12.836019 M	white15.35
green!15HalfWaveUNet	13.452659 M	white12.53
green!15HalfWaveUnet2	12.788763 M	white9.6
green!20UnetNaive	12.721059 M	white9.81
yellow!25waveUnet_idwt	12.878891 M	white24.56
red!15Unet_Res	8.743131 M	white32.25
red!15WaveUNet_Res	7.779147 M	white32.25
red!25Unet_Res4	12.272715 M	white50.26
red!15Unet_Res4_att	13.012875 M	white32.25
yellow!20Unet_Fixed	13.809033 M	white23.63
orange!30UnetDWT2	11.514291 M	white17.09
orange!30UnetDWT2_bicube	11.514291 M	white32.25
green!10Att2Wave	13.751107 M	white6.82
green!15Att2Wave64	13.907827 M	white7.337
green!5Att2Wave_4	19.822083 M	white4.457
red!20Att2Wave_4Wave	12.719499 M	white49.75
orange!30UnetDWT2_att2wave	11.404035 M	white34.62
green!15Att2Wave2	13.850243 M	white10.28
green!15Att2Wave_pixshuf	13.708867 M	white12.17
green!20InternImage	13.87 M	white10.87
red!25LiteVAE	>14 M	white32.53

- **Att2Wave** consistently achieved the best trade-off between FID and parameter count, outperforming most U-NET-based models while removing all attention mechanisms. Its use of WaveBlocks proved both computationally efficient and generatively powerful.

- **WaveUNet** and its variants offered strong middle-ground performance, particularly HalfWaveUnet2, which maintained low FID with fewer parameters.

- **InternImage** models, when used to replace UNET residual blocks, demonstrated competitive results but did not surpass the efficiency of Att2Wave.

- The **LiteVAE** latent-space diffusion model showed limitations in preserving fine spatial details, resulting in higher FID.

- Surprisingly, even the stripped-down **UnetNaive** variant performed reasonably well, suggesting that complex attention mechanisms and deep hierarchies may not be essential for acceptable generative performance.

We further performed an ablation study on architectural variants based on InternImage-style building blocks, isolating the effects of skip connections, sampling methods, and block types.

TABLE 2.2: Experimental Results on CIFAR-10: Across InternImage Variants

2gray!5white

gray!25 **Skip connection**	**Downsampling**	**Upsampling**	**Parameters**	white**FID**
green!20Yes	Pooling	Bilinear	13.52 M	white11.88
green!20Yes	Pooling	Bicubic	13.52 M	white11.75
orange!20Yes	Pooling	PixelShuffle	12.83 M	white12.38
green!10Yes	Pooling	WaveMixSR-V2	14.87 M	white11.37
green!10Yes	Pooling	CNN stride 2	14.87 M	white11.39
green!10Yes	WaveMix	WaveMixSR-V2	16.32 M	white11.11
green!10Yes	WaveMix	CNN stride 2	14.98 M	white11.10
green!15Yes	Transpose CNN	WaveMixSR-V2	15.11 M	white10.86
green!15Yes	Transpose CNN	CNN stride 2	13.87 M	white10.87
red!20No	Transpose CNN	CNN stride 2	12.69 M	white15.76

Key insights from this analysis include:

- The presence of **skip connections** has a significant impact on generative quality. Removing them leads to sharp degradation in FID.

- Parametric upsampling techniques such as **CNN stride-2** and **WaveMixSR-V2** consistently outperform non-parametric methods like bilinear, bicubic, or pixel shuffle. However, they come at the cost of increased model size.

- Between WaveMix and CNN sampling strategies, CNN-based upsampling was chosen due to its favorable balance between performance and parameter efficiency.

2.5.4 Cross-Dataset Evaluation

Table 2.3: Experimental Results on MNIST: Across Model Variants

	gray!25 **Model**	**Parameters**	white**FID**
	yellow!30UNET	13.796545 M	white60.43
	red!25WavemixSR	12.987076 M	white>100
	yellow!30WaveUNET	12.832561 M	white58.147
2gray!5white	yellow!30HalfWaveUnet	13.449241 M	white60.951
	green!20HalfWaveUnet2	12.785305 M	white56.691
	yellow!30UnetNaive	12.717601 M	white61
	yellow!30WaveUnet_idwt	12.875433 M	white59.42
	green!25Att2Wave	8.172353 M	white54.516

Table 2.4: Experimental Results on TinyImageNet: Across Model Variants

	gray!25 **Model**	**Parameters**	white**FID**
	green!20UNET	13.800003 M	white37.293
	green!20Att2Wave	13.751107 M	white37.293
	yellow!25Att2Wave_4	19.822083 M	white66.6923
2gray!5white	yellow!20Att2Wave64	13.907827 M	white62.351
	yellow!30Att2Wave_pixshuf	13.708867 M	white73.468
	orange!40UnetDWT2	11.514291 M	white167.274
	orange!30Att2Wave2	13.850243 M	white112.35
	red!25waveUnetIdwt	12.878891 M	whiteHigh

The models were further tested on Tiny ImageNet, CelebA, and MNIST. Key observations include:

- Att2Wave consistently achieved comparable or superior FID scores across all datasets, showing that the removal of attention did not harm generalization and that wavelet-based blocks could effectively replace them.

- In Tiny ImageNet, which contains high intra-class variability and fine texture, WaveUNet_IDWT performed particularly well, leveraging wavelet-based spatial reconstruction.
- On CelebA, models that preserved higher frequency components (like Att2Wave and WaveUNet_IDWT) generated sharper and more identity-preserving faces.
- For MNIST, even the most minimalistic models (like UnetNaive) achieved low FID scores, suggesting that simpler tasks benefit less from architectural complexity.

These results affirm that the proposed architectures are scalable and robust across varying data domains.

2.5.5 Scale Evaluation on Celeb-A

The CelebA dataset, consisting of over 200,000 celebrity face images, provides a useful benchmark for evaluating the generative model's ability to preserve structural features like identity, symmetry, and local texture. We evaluate both the baseline UNET and the Att2Wave model, which emerged as the best-performing architecture across other datasets in this thesis.

TABLE 2.5: FID scores on CelebA
Att2Wave and UNET perform identically in terms of FID after 5 epochs

gray!25 **Model**	**Parameters**	**FID**
UNET	13.800003 M	36.16
Att2Wave	13.751107 M	36.16

Despite the complexity of the dataset and higher resolution (128×128), both models achieved an FID of 36.168 after 5 epochs. However, Att2Wave reached this performance using approximately 41% fewer parameters (e.g., 8.17M vs. 13.79M on

Table 2.6: U-NET and Att2Wave Models' Comparison across datasets
Att2Wave consistently uses fewer parameters and achieves comparable or better performance

gray!25 **Model**	**MNIST**	**CIFAR-10**	**TinyImageNet**	**CelebA-128**
FID Score				
UNET	60.43	7.29	37.293	36.168
Att2Wave	54.516	6.82	37.293	36.168
Parameter Count (Millions)				
UNET	13.796545	13.800003	13.800003	13.800003
Att2Wave	8.172353	13.751107	13.751107	13.751107

MNIST), highlighting its superior parameter efficiency. The consistent performance across MNIST, CIFAR-10, TinyImageNet, and now CelebA confirms Att2Wave's generalization ability and robustness.

These findings strengthen the case for attention-free diffusion models, particularly those based on wavelet-driven spatial interaction. Unlike UNET, which relies on explicit self-attention mechanisms to capture dependencies, Att2Wave uses lightweight WaveBlocks to achieve similar results at a lower computational cost. This supports recent trends in generative modeling that favor simplified, non-attention-based architectures for scalability Rombach et al. (2022); Wang et al. (2022).

In visual samples, both architectures generated identity-consistent, high-quality faces, but Att2Wave models exhibited slightly better detail recovery around facial features like the eyes and mouth. These results suggest that wavelet-based spatial mixing may be particularly effective for domains with subtle texture and semantic constraints, such as face synthesis.

2.5.6 Qualitative Analysis

Beyond quantitative metrics, we visually assessed the generative outputs of various architectures. Models like Att2Wave and WaveUNet_IDWT produced sharper edges and better global structure, especially on CelebA and TinyImageNet. In contrast, models operating purely in latent space (LiteVAE) sometimes exhibited loss of fine textures, though overall content and pose remained consistent.

2.5.7 Impact of Step Block Complexity

One interesting observation from our experiments was that the complexity of the step block and CNN layers at the beginning and end of the model had little to no significant impact on the FID score. Even simple CNNs used to map the number of input and output channels were sufficient to achieve strong performance.

This reinforces the notion that core architectural choices such as resolution manipulation strategy and spatial mixing mechanisms have a far greater influence on generative performance than the depth or expressiveness of early/late convolutional layers.

2.5.8 LiteVAE Performance and Latent-Space Trade-offs

The LiteVAE + diffusion pipeline provides a compelling case for reducing dimensionality before the diffusion process. The encoder and decoder of LiteVAE were trained using a combination of six loss terms, including L1, perceptual (VGG), adversarial, KL divergence, and high-frequency losses.

Once trained, the diffusion model operates on the latent code instead of the full image, and the decoder reconstructs the final output. This two-stage approach resulted in significantly reduced inference time, especially on larger images, acceptable FID scores, though slightly worse than pixel-space models and less parameter overhead, due to smaller model input sizes.
LiteVAE is particularly attractive for real-time or embedded scenarios where inference speed is critical and small degradations in quality are tolerable.

2.6 Summary

This thesis set out to address one of the core challenges in the practical deployment of diffusion models: their computational inefficiency and large parameter footprint. While diffusion models have demonstrated remarkable success in image generation tasks, their standard implementations—typically involving attention-heavy U-NETs—make them unsuitable for resource-constrained environments. Through a systematic exploration of resource-efficient architectural alternatives, we aimed to answer the central question: Can we build diffusion models that are significantly lighter, faster, and more scalable, without compromising on output quality?

2.6.1 Motivation Recap

Diffusion models are inherently iterative and require passing a noisy image through a deep neural network for hundreds of steps. When this neural network is a large, attention-based U-NET, the result is high memory usage, long inference time, and limited deployability on edge or mobile devices. With applications of generative modeling expanding into real-time imaging, medical diagnostics, and creative tools, there is a clear and growing need for more efficient generative architectures.

Our work was driven by this motivation—to rethink architectural design choices in diffusion models, leveraging ideas from wavelet theory, convolutional geometry, and latent space compression to significantly reduce the computational load.

2.6.2 Approaches Explored

We proposed and evaluated several families of architectures, each targeting a different bottleneck in diffusion model design:

1. WaveUNet and its Variants

   Replaced standard upsampling and downsampling blocks with wavelet-based resolution manipulation, achieving parameter-efficient multi-scale representations. Multiple variants explored different trade-offs between learnability and simplicity.

2. Att2Wave

   Eliminated the use of attention mechanisms by replacing them with Wave-Blocks, which provide implicit multi-scale interactions at a fraction of the cost. Att2Wave emerged as one of the most effective designs, achieving the best trade-off between FID and efficiency.

3. InternImage (DCNv3-based)

   Integrated deformable convolutional networks to replace standard residual blocks, providing adaptive spatial aggregation with fewer parameters. While powerful, InternImage-based designs did not always outperform wavelet-based models in terms of efficiency.

4. LiteVAE

   Shifted diffusion from image space to latent space, reducing the size and complexity of the input to the diffusion model. This enabled faster inference and compact diffusion, especially beneficial in constrained environments.

Each of these architectures was subjected to detailed empirical evaluation on diverse datasets such as CIFAR-10, Tiny ImageNet, CelebA, and MNIST, providing a comprehensive view of their capabilities and limitations.

2.6.3 Experimentation Insights

From our results, several key insights emerged:

- Wavelet-based architectures offer an excellent alternative to attention in diffusion models. Not only do they significantly reduce parameter counts, but they also yield comparable or improved generative quality (measured via FID).
- Naïve architectures—such as average pooling for downsampling and interpolation for upsampling—performed better than expected. This challenges the assumption that high complexity is necessary for high-quality generation, particularly in simple domains like MNIST.
- Att2Wave outperformed all other models in the parameter-to-performance trade-off, showing that attention mechanisms can be safely removed under certain design constraints.
- InternImage blocks provide a viable middle ground, combining inductive biases from CNNs with flexibility from transformers. They offer better accuracy than naïve models and better scalability than full attention.

- Latent-space diffusion (via LiteVAE) is a promising direction for high-speed applications, although it requires more careful training and tends to sacrifice some sharpness or high-frequency detail.

The complexity of early and late layers (step/CNN blocks) was found to have minimal impact on overall FID, further emphasizing the importance of core architectural structure over depth.

2.6.4 Overall Contribution

This work contributes to the growing body of literature on efficient generative modeling in the following ways:

- Proposes and validates wavelet-based architectural alternatives to traditional diffusion components.
- Demonstrates that attention-free models can rival state-of-the-art methods in performance.
- Provides extensive empirical benchmarks across multiple datasets, highlighting the practical relevance of each model variant.
- Bridges the gap between theoretical efficiency and empirical performance, paving the way for diffusion models in low-power and real-time settings.

2.6.5 Closing Thoughts

Our exploration of resource-efficient diffusion architectures reveals that with thoughtful design, it is possible to drastically reduce the computational requirements of generative models without a proportional loss in quality. The key lies not in replicating the complexity of traditional models, but in leveraging structured transformations such as wavelets, deformable convolutions, and latent space projections that capture essential features in more compact forms.

As generative models continue to evolve and find new applications, this work serves as a step toward sustainable, deployable, and accessible diffusion-based generative AI.

Looking forward, integrating these architectural ideas with newer sampling schemes (such as consistency models (Song et al., 2023) and shortcut models (Frans et al., 2025)) and learning-to-learn paradigms (e.g., diffusion adapters) can further enhance practicality. Evaluating these models in real-world tasks like conditional image synthesis, video generation, and controllable diffusion is a promising next step.

Appendix A

Evidence lower Bound : ELBO

Evidence Lower Bound or ELBO is used as a training objective in a lot of variational models. It provides a lower bound on the true but intractible function we intend to optimize.

Mathematically, let $x(x)$ denote the likelihood of the observed quantity x, which we intend to maximize. Let $p(z, x)$ be the joint likelihood of variables in some latent space, z and the observed x. A way to obtain the true likelihood, thus, is to marginalize out the latent variable from the joint probability function :

$$p(x) = \int p(z, x) \, dz$$

However, this is intractable for complex models. Another way is to take into use the chain rules of probability :

$$p(x) = \frac{p(z, x}{p(z|x}$$

This requires the knowledge of $p(z|x)$. We approximate this using another function, $q_\phi(z|x)$, where ϕ are trainable parameters. We can mathematically write :

$$\begin{aligned}
\log p(x) &= \log \int p(z,x)dz \\
&= \log \int \frac{p(z,x)q_\phi(z|x)}{q_\phi(z|x)}dz \\
&= \log \mathbb{E}_{q_\phi(z|x)}\left[\frac{p(z,x)}{q_\phi(z|x)}\right] \\
&\geq \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(z,x)}{q_\phi(z|x)}\right]
\end{aligned}$$

The term derived in the RHS is known as **Evidence Lower Bound**. In fact, it can be shown that the LHS is greater than the RHS in the above by precisely the KL Divergence between $p(z|x)$ and $q_\phi(z|x)$. Thus, the bound is tight.

A.1 ELBO for VAEs

In the standard formulation of the Variational Autoencoder (VAE), the objective is to directly maximize the Evidence Lower Bound (ELBO). The term "autoencoder" is used because the model resembles a traditional autoencoder, where the input data is encoded into a compressed form and then reconstructed. To clarify this connection, let's break down the ELBO term in more detail:

$$\begin{aligned}
\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(z,x)}{q_\phi(z|x)}\right] &= \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x|z)p(z)}{q_\phi(z|x)}\right] \\
&= \mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(z|x)\right] + \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(z)}{q_\phi(z|x)}\right] \\
&= \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(z|x)\right]}_{\text{reconstruction term}} - \underbrace{\mathcal{D}_{KL}(q_\phi(z|x) \mid\mid p(z))}_{\text{KL term}}
\end{aligned}$$

Here, p_θ can be treated as an "encoder", and q_ϕ as the "decoder", the respective subscripts denoting parameters to be trained.

Thus, for the case of VAEs, the ELBO neatly splits into a reconstruction loss and a KL loss, each of which has a very intuitive role in the formulation (see section 1.1.1)

References

Balasubramanian, K., Chewi, S., Erdogdu, M., Salim, A., Zhang, M., 2022. Towards a theory of non-log-concave sampling: First-order stationarity guarantees for langevin monte carlo. doi:10.48550/arXiv.2202.05214.

Bao, F., Li, C., Sun, J., Zhu, J., Zhang, B., 2022. Estimating the optimal covariance with imperfect mean in diffusion probabilistic models. arXiv preprint arXiv:2206.07309 .

Betker, J., Goh, G., Jing, L., Brooks, T., Wang, J., Li, L., Ouyang, L., Zhuang, J., Lee, J., Guo, Y., et al., 2023. Improving image generation with better captions. Computer Science. https://cdn. openai. com/papers/dall-e-3. pdf 2, 8.

Brosse, N., Durmus, A., Moulines, , Pereyra, M., 2017. Sampling from a log-concave distribution with compact support with proximal langevin monte carlo, in: Kale, S., Shamir, O. (Eds.), Proceedings of the 2017 Conference on Learning Theory, PMLR. pp. 319–342. URL: https://proceedings.mlr.press/v65/brosse17a.html.

Chen, S., Guo, W., 2023. Auto-encoders in deep learning—a review with new perspectives. Mathematics 11, 1777. doi:10.3390/math11081777.

Croitoru, F.A., Hondru, V., Ionescu, R.T., Shah, M., 2023. Diffusion models in vision: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence PP, 1–20. doi:10.1109/TPAMI.2023.3261988.

Dai, J., Qi, H., Xiong, Y., Li, Y., Zhang, G., Hu, H., Wei, Y., 2017. Deformable convolutional networks. URL: https://arxiv.org/abs/1703.06211, arXiv:1703.06211.

Dalalyan, A.S., 2014. Theoretical guarantees for approximate sampling from smooth and log-concave densities. Working Papers 2014-45. Center for Research in Economics and Statistics. URL: https://ideas.repec.org/p/crs/wpaper/2014-45.html.

Datar, M., Immorlica, N., Indyk, P., Mirrokni, V.S., 2004. Locality-sensitive hashing scheme based on p-stable distributions, in: Proceedings of the Twentieth Annual Symposium on Computational Geometry, Association for Computing Machinery, New York, NY, USA. p. 253–262. URL: https://doi.org/10.1145/997817.997857, doi:10.1145/997817.997857.

Erdogdu, M.A., Hosseinzadeh, R., 2020. On the convergence of langevin monte carlo: The interplay between tail growth and smoothness, in: Annual Conference Computational Learning Theory. URL: https://api.semanticscholar.org/CorpusID:218900990.

Frans, K., Hafner, D., Levine, S., Abbeel, P., 2025. One step diffusion via shortcut models. URL: https://arxiv.org/abs/2410.12557, arXiv:2410.12557.

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y., 2014. Generative adversarial networks. Advances in Neural Information Processing Systems 3. doi:10.1145/3422622.

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y., 2020. Generative adversarial networks. Communications of the ACM 63, 139–144.

Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S., 2017. Gans trained by a two time-scale update rule converge to a local nash equilibrium, in: Proceedings of the 31st International Conference on Neural Information Processing Systems, Curran Associates Inc., Red Hook, NY, USA. p. 6629–6640.

Ho, J., Jain, A., Abbeel, P., 2020. Denoising diffusion probabilistic models. Advances in neural information processing systems 33, 6840–6851.

Jeevan, P., Nixon, N., Sethi, A., 2024a. Normalizing flow-based metric for image generation. URL: https://arxiv.org/abs/2410.02004, arXiv:2410.02004.

Jeevan, P., Nixon, N., Sethi, A., 2024b. Wavemixsr-v2: Enhancing super-resolution with higher efficiency. URL: https://arxiv.org/abs/2409.10582, arXiv:2409.10582.

Jeevan, P., Srinidhi, A., Prathiba, P., Sethi, A., 2023. Wavemixsr: A resource-efficient neural network for image super-resolution. URL: https://arxiv.org/abs/2307.00430, arXiv:2307.00430.

Jeevan, P., Viswanathan, K., S, A.A., Sethi, A., 2024c. Wavemix: A resource-efficient neural network for image analysis. URL: https://arxiv.org/abs/2205.14375, arXiv:2205.14375.

Kingma, D.P., Salimans, T., Jozefowicz, R., Chen, X., Sutskever, I., Welling, M., 2016. Improved variational inference with inverse autoregressive flow. Advances in neural information processing systems 29.

Kingma, D.P., Welling, M., 2013. Auto-encoding variational bayes. arXiv preprint arXiv:1312.6114 .

Kingma, D.P., Welling, M., 2022. Auto-encoding variational bayes. URL: https://arxiv.org/abs/1312.6114, arXiv:1312.6114.

Liu, L., Chen, Y., Li, C., Yu, C., Wang, L.C., 2020. Understanding the role of normalization in deep learning, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 438–451.

Luo, C., 2022. Understanding diffusion models: A unified perspective. ArXiv abs/2208.11970. URL: https://api.semanticscholar.org/CorpusID:251799923.

McInnes, L., Healy, J., Saul, N., Großberger, L., 2018. Umap: Uniform manifold approximation and projection. Journal of Open Source Software 3, 861. URL: https://doi.org/10.21105/joss.00861, doi:10.21105/joss.00861.

Mousavi-Hosseini, A., Farghly, T.K., He, Y., Balasubramanian, K., Erdogdu, M.A., 2023. Towards a complete analysis of langevin monte carlo: Beyond poincaré inequality, in: Neu, G., Rosasco, L. (Eds.), Proceedings of Thirty Sixth Conference on Learning Theory, PMLR. pp. 1–35. URL: https://proceedings.mlr.press/v195/mousavi-hosseini23a.html.

Nichol, A.Q., Dhariwal, P., 2021. Improved denoising diffusion probabilistic models, in: International conference on machine learning, PMLR. pp. 8162–8171.

Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I., 2021. Learning transferable visual models from natural language supervision, in: International

Conference on Machine Learning. URL: https://api.semanticscholar.org/CorpusID:231591445.

Roberts, G.O., Tweedie, R.L., 1996. Exponential convergence of Langevin distributions and their discrete approximations. Bernoulli 2, 341 – 363.

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B., 2022. High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 10684–10695.

Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation, pp. 234–241. doi:10.1007/978-3-319-24574-4_28.

Sadat, S., Buhmann, J., Bradley, D., Hilliges, O., Weber, R.M., 2025. Litevae: Lightweight and efficient variational autoencoders for latent diffusion models. URL: https://arxiv.org/abs/2405.14477, arXiv:2405.14477.

Sønderby, C.K., Raiko, T., Maaløe, L., Sønderby, S.K., Winther, O., 2016. Ladder variational autoencoders. Advances in neural information processing systems 29.

Song, J., Meng, C., Ermon, S., 2021. Denoising diffusion implicit models, in: International Conference on Learning Representations. URL: https://openreview.net/forum?id=St1giarCHLP.

Song, J., Meng, C., Ermon, S., 2022. Denoising diffusion implicit models. URL: https://arxiv.org/abs/2010.02502, arXiv:2010.02502.

Song, Y., Meng, C., Ermon, S., 2023. Consistency models, in: ICML.

Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., Poole, B., 2020. Score-based generative modeling through stochastic differential equations. CoRR abs/2011.13456. URL: https://arxiv.org/abs/2011.13456, arXiv:2011.13456.

Stein, G., Cresswell, J.C., Hosseinzadeh, R., Sui, Y., Ross, B.L., Villecroze, V., Liu, Z., Caterini, A.L., Taylor, J.E.T., Loaiza-Ganem, G., 2023. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models, in: Proceedings of the 37th International Conference on Neural Information Processing Systems, Curran Associates Inc., Red Hook, NY, USA.

Wang, H., 2024. Comparative analysis of gans and diffusion models in image generation. Highlights in Science, Engineering and Technology 120, 59–66. doi:10.54097/9gba9v27.

Wang, W., Dai, J., Chen, Z., Huang, Z., Li, Z., Zhu, X., Hu, X., Lu, T., Lu, L., Li, H., Wang, X., Qiao, Y., 2023. Internimage: Exploring large-scale vision foundation models with deformable convolutions, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 14408–14419. doi:10.1109/CVPR52729.2023.01385.

Wang, Y., Li, Z., et al., 2022. Internimage: Exploring plain vision backbone for object detection and segmentation. arXiv preprint arXiv:2211.05778 .

Yaqoob, M.H., Ahmad, Z., et al., 2022. Wavemix: A resource-efficient neural network for image analysis. arXiv preprint arXiv:2206.09765 .

Zhang, B., Titov, I., Sennrich, R., 2022. Root mean square layer normalization, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 23118–23130.

Zhu, X., Hu, H., Lin, S., Dai, J., 2019. Deformable convnets v2: More deformable, better results, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 9300–9308. doi:10.1109/CVPR.2019.00953.